AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Course Title

INTRODUCTION TO DATABASE

Summer, 2023-2024
Section: I

TITLE

SuperTechIT: Super Shop Database Management System

Supervised By

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Table Creation (DDL Operations)

StudentID1: 23-51788-2 Name: Sanjida Affrin Bristi	StudentID3: 23-51756-2 Name: Shreya Chakraborty
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CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

Create Tables

1. Employee Table

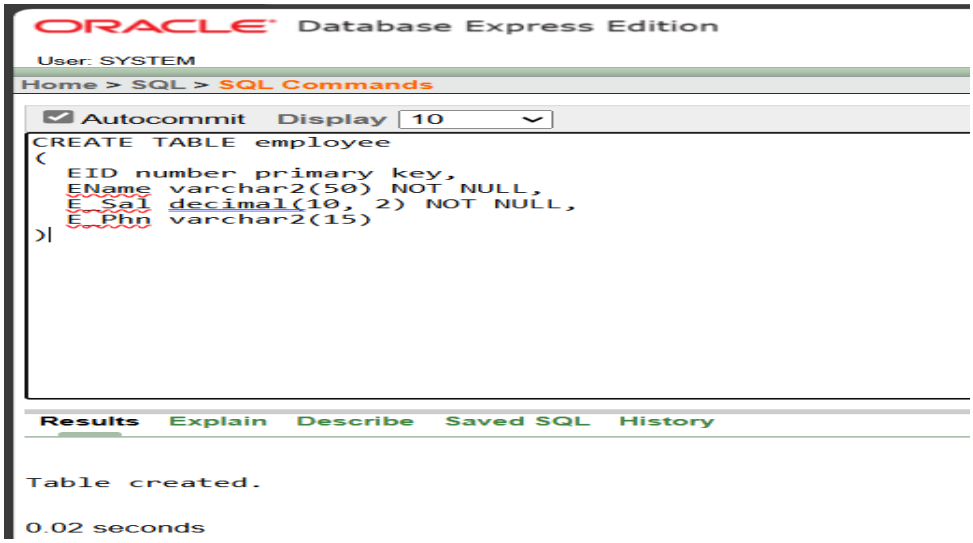


Fig-1: Create employee table

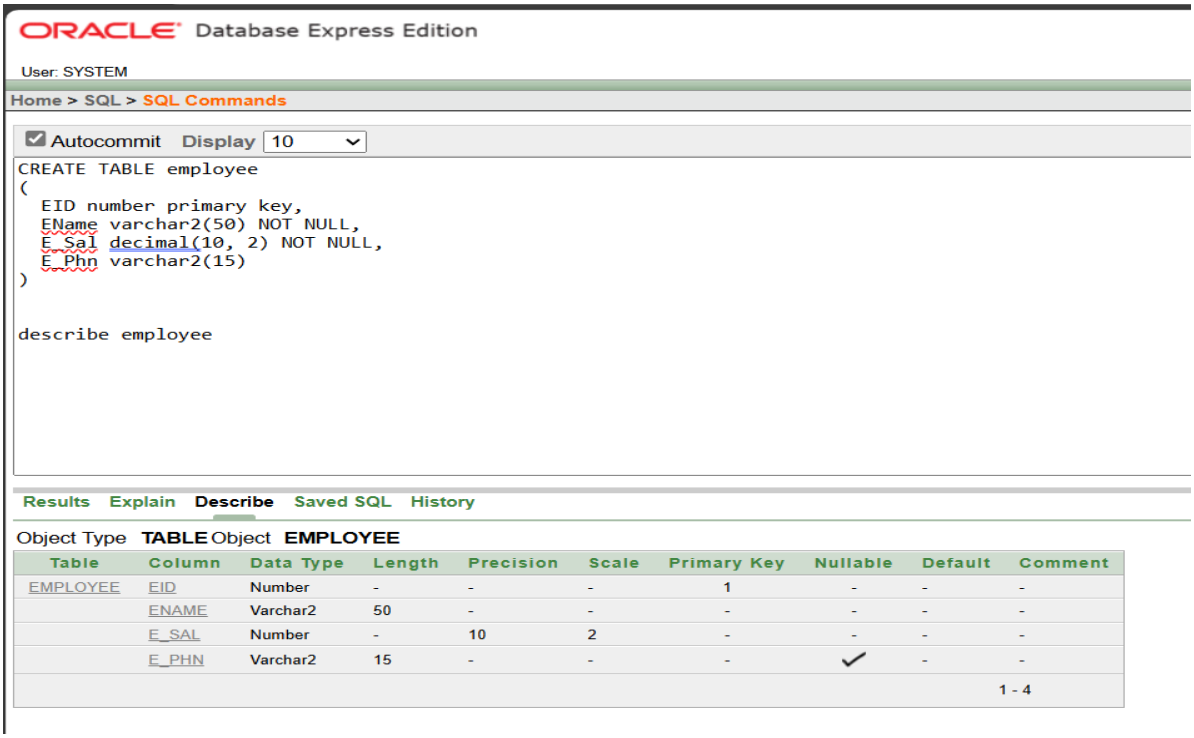


Fig-2: Description of the employee table

2. Create Customer Table

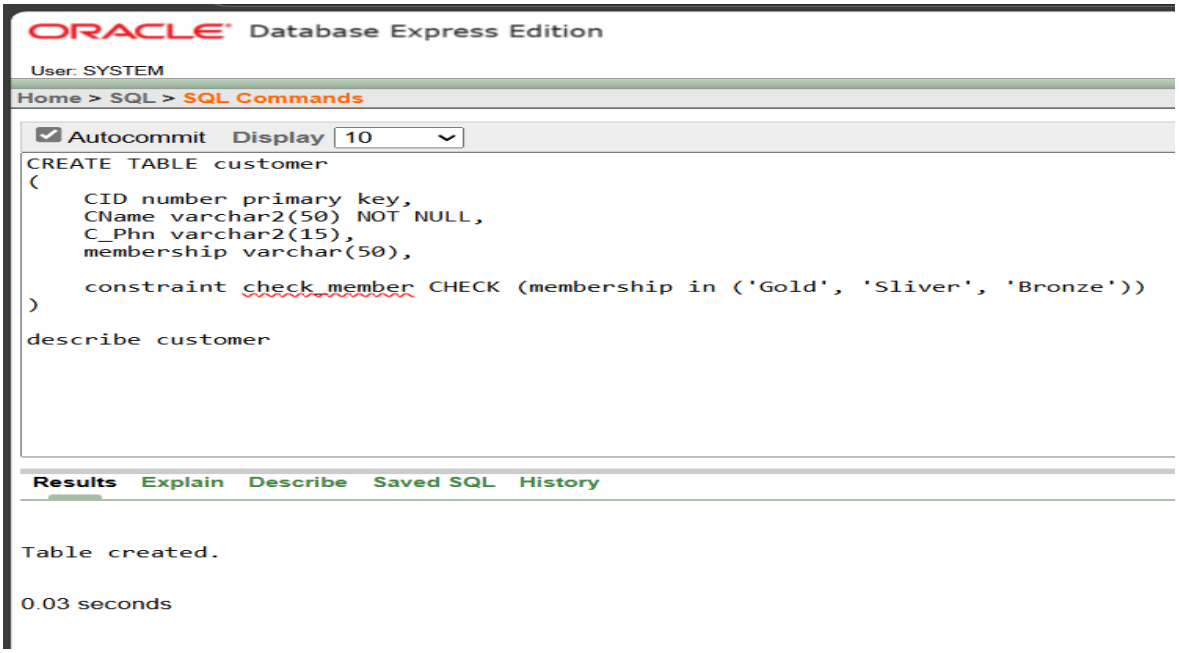


Fig-3: Create customer table

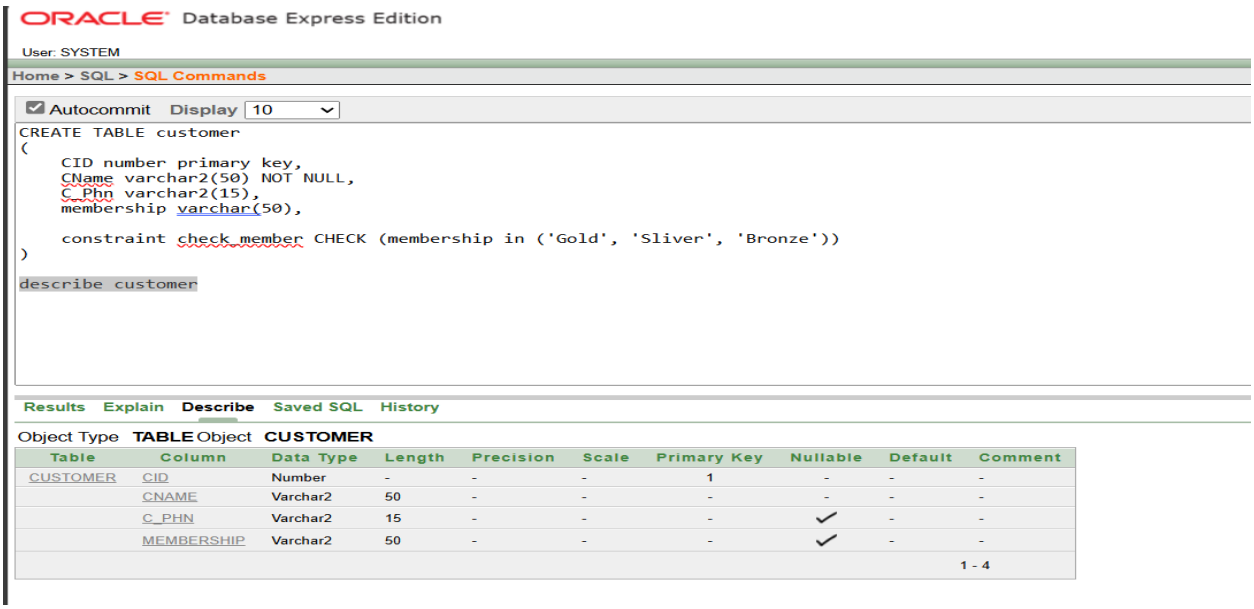


Fig-4: Description of the customer table

3. Product Table

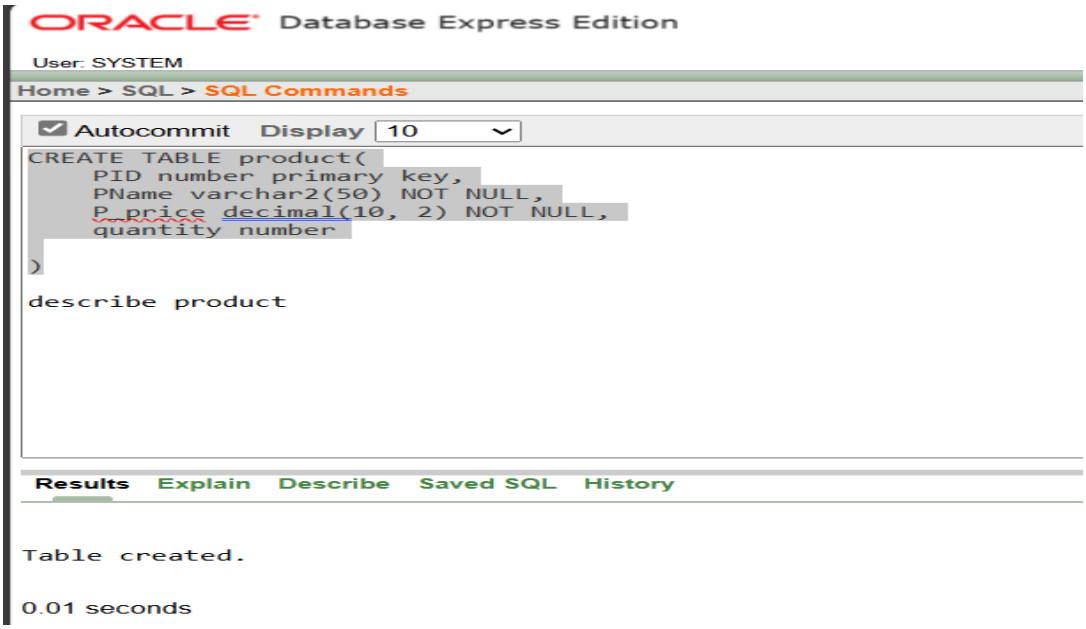


Fig-5: Create product table

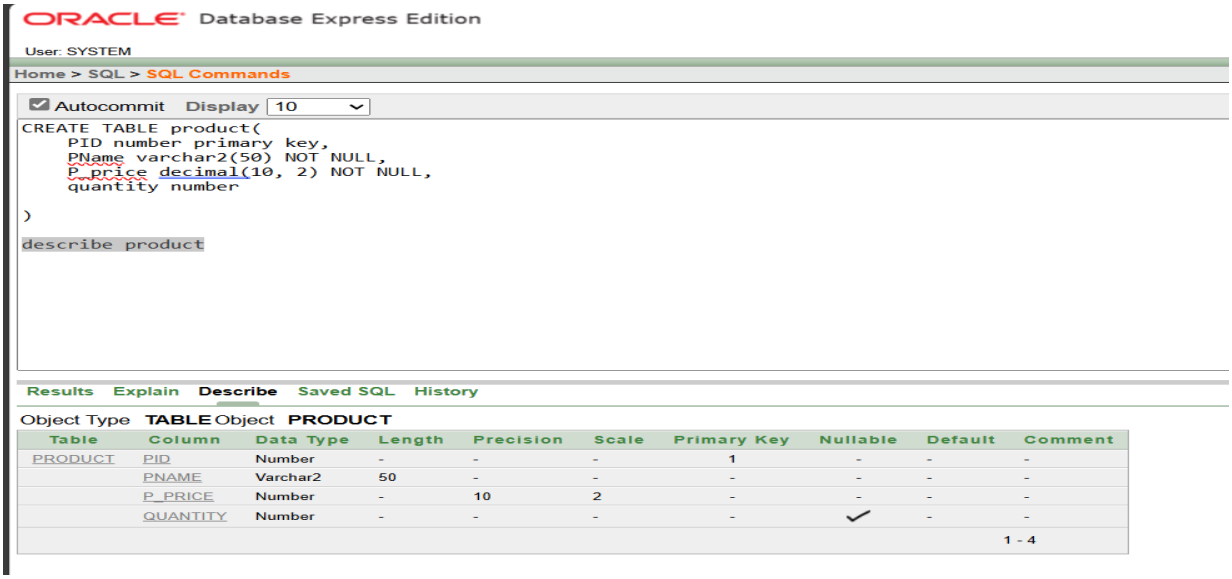


Fig-6: Description of the product table

4. Orders Table

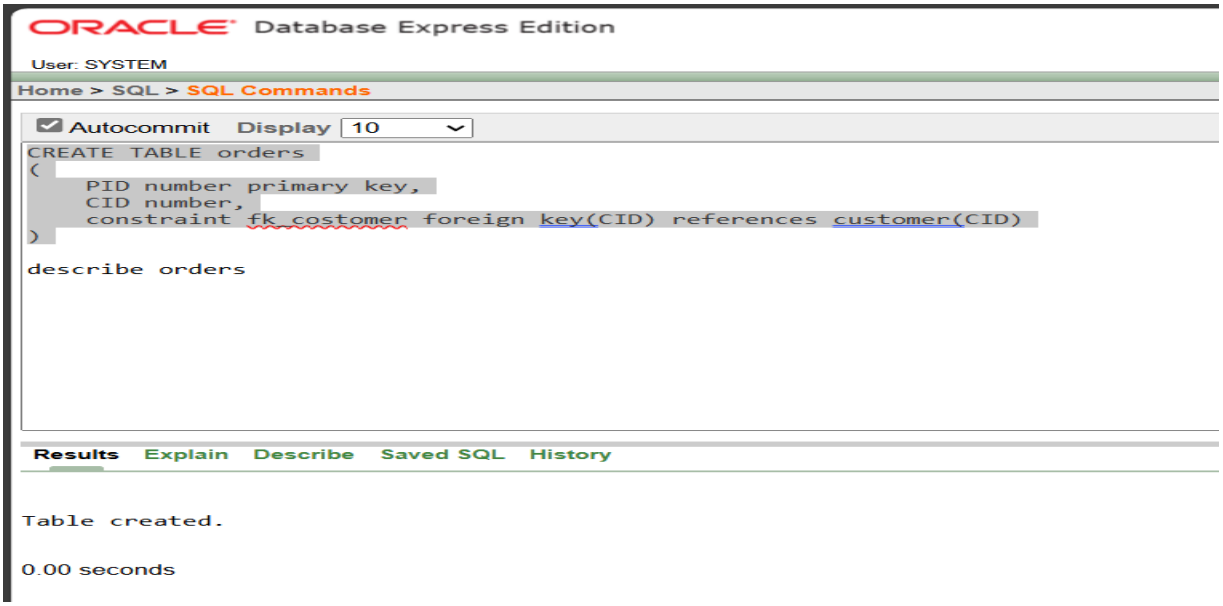


Fig-7: Create order table

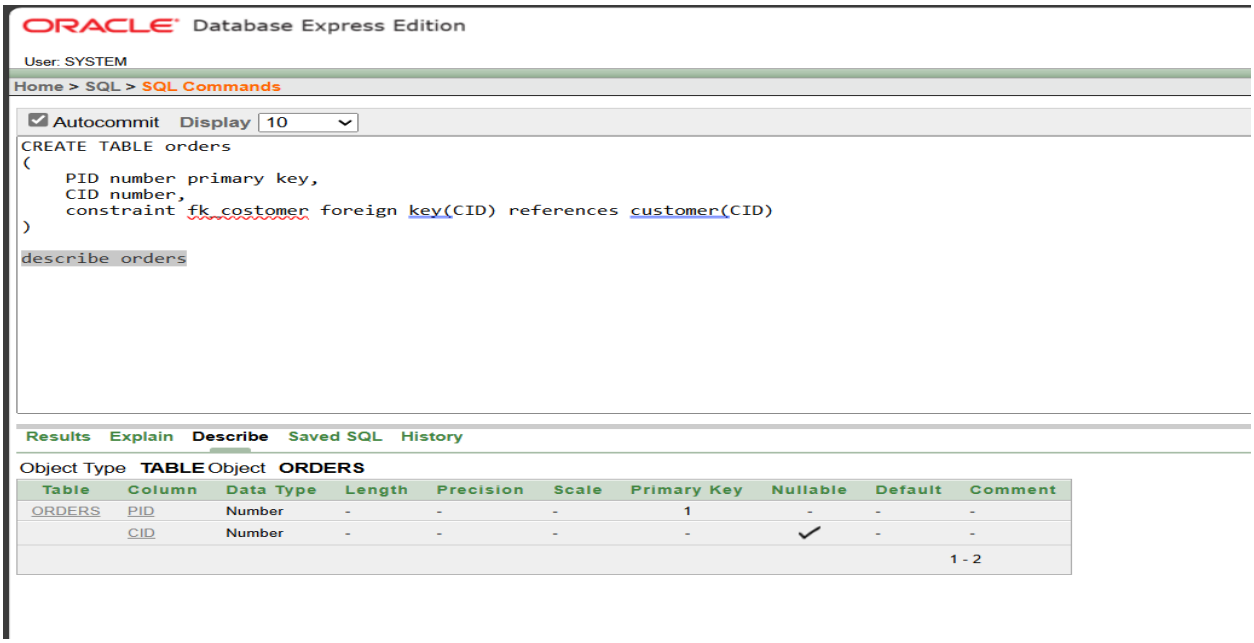


Fig-8: Description of the orders table

5. Handle Table

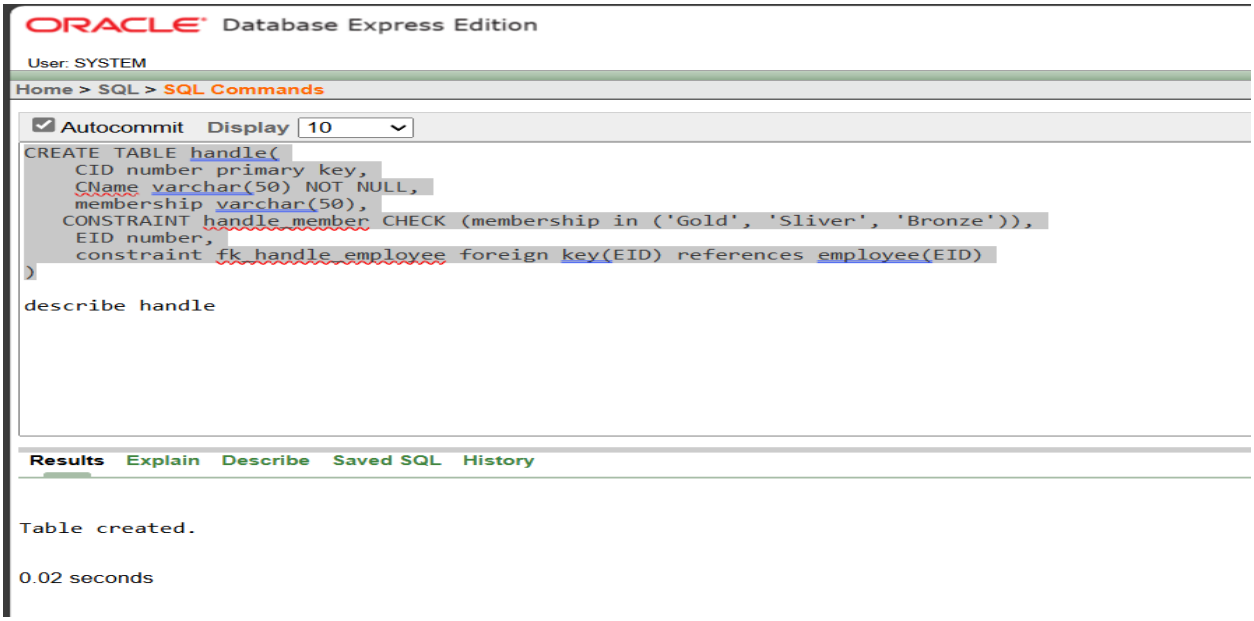


Fig-9: Create handle table

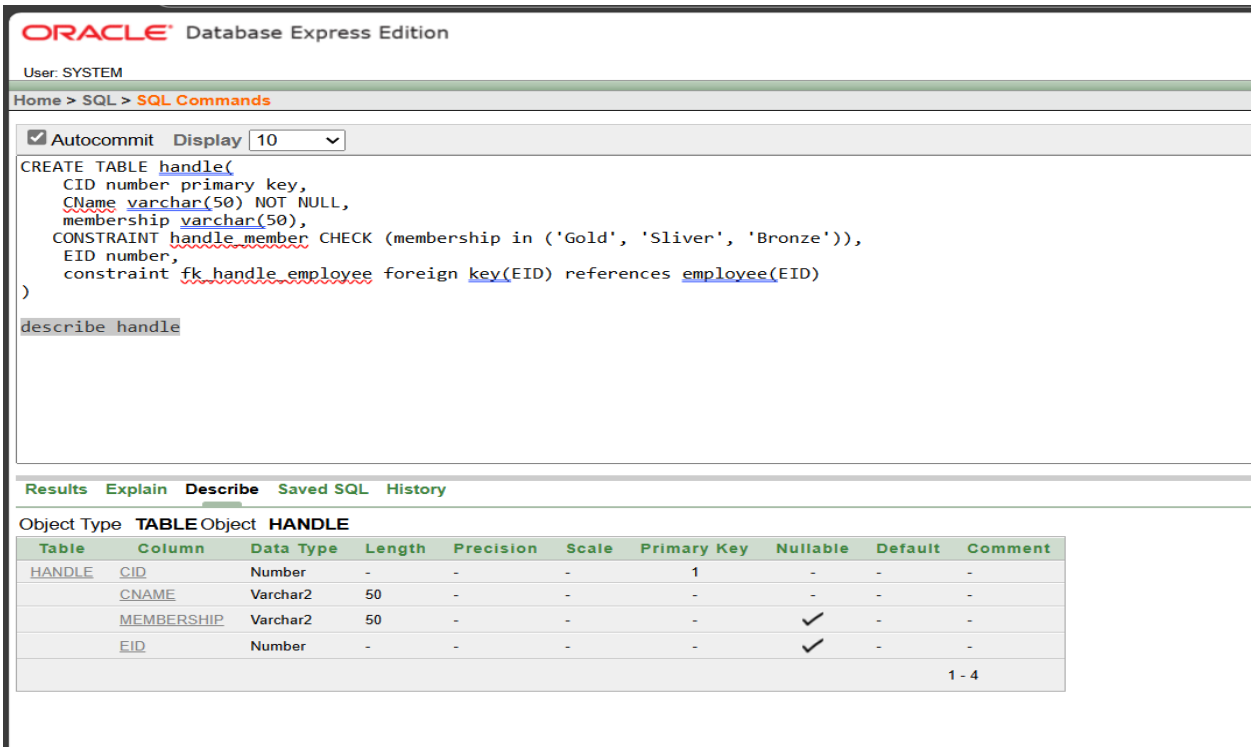


Fig-10: Description of the handle table

6. Restore Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 10

```
CREATE TABLE restore
(
  PID number primary key,
  EID number,
  constraint fk_restore foreign key(EID) references employee(EID)
)

describe store
```

Results Explain Describe Saved SQL History

```
Table created.

0.02 seconds
```

Fig-11: Create restore table

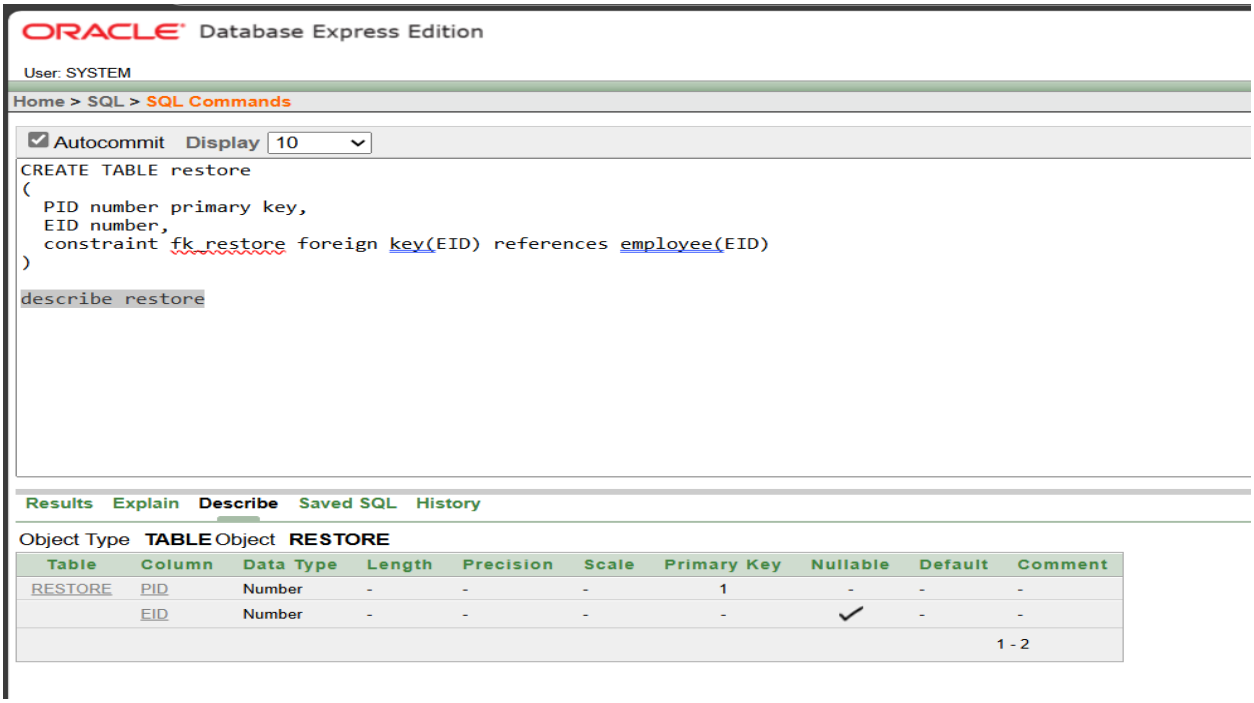


Fig-12: Description of the restore table

7. Company Table

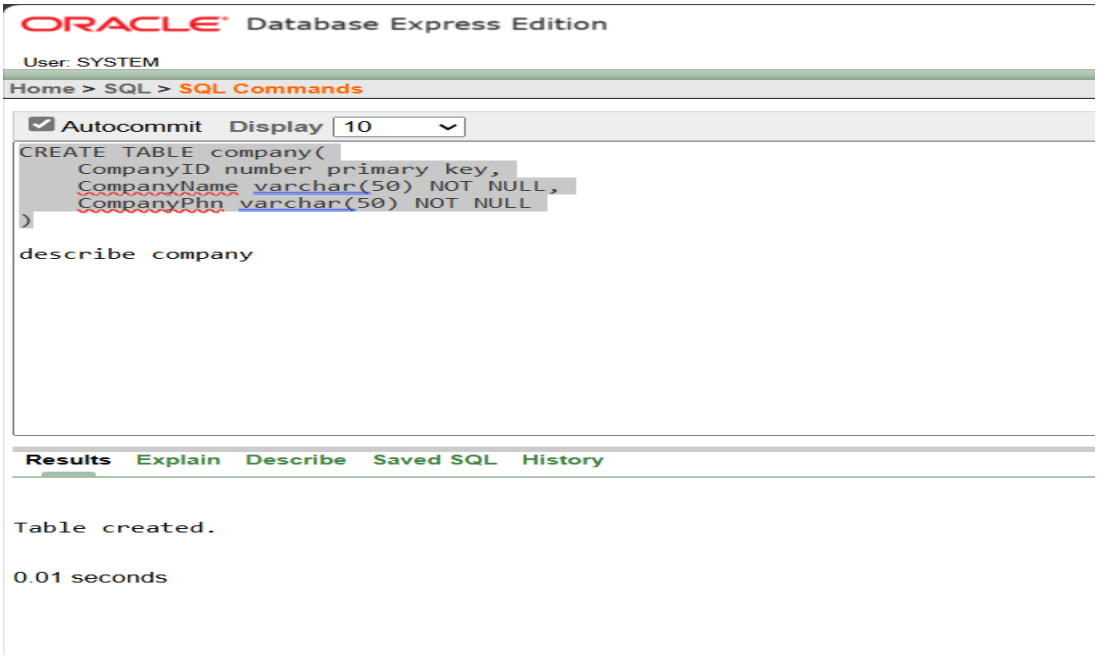


Fig-13: Create company table

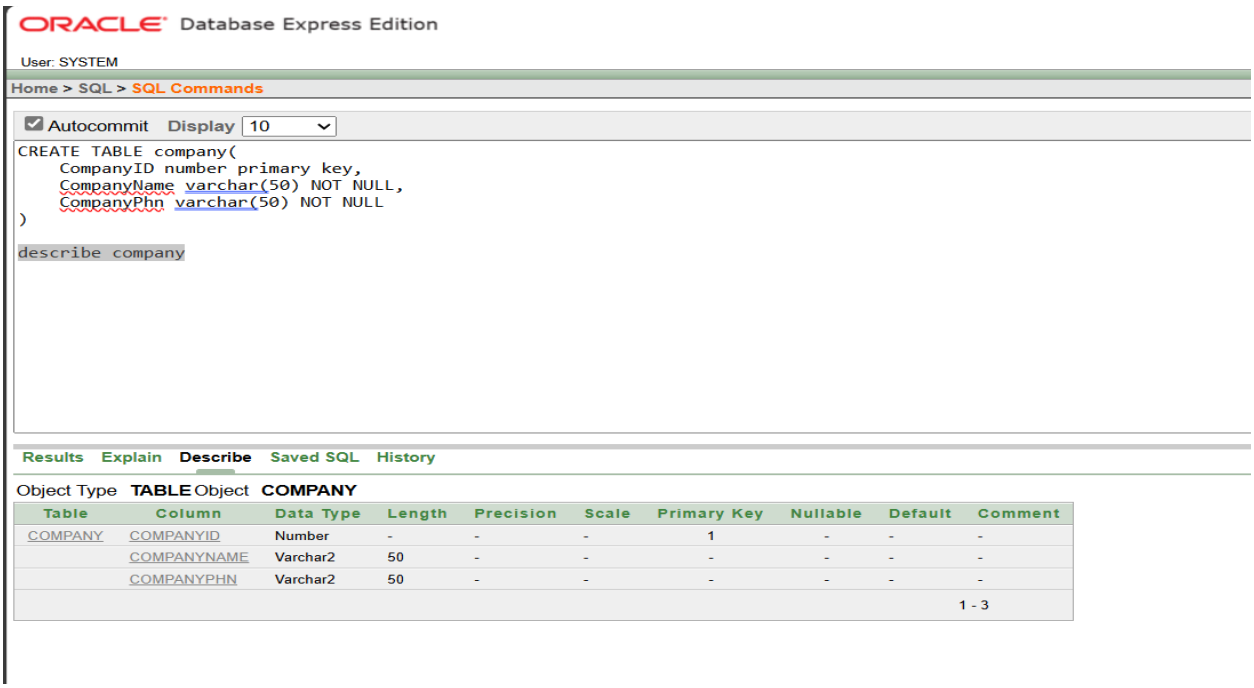


Fig-14: Description of the company table

8. Supply Table

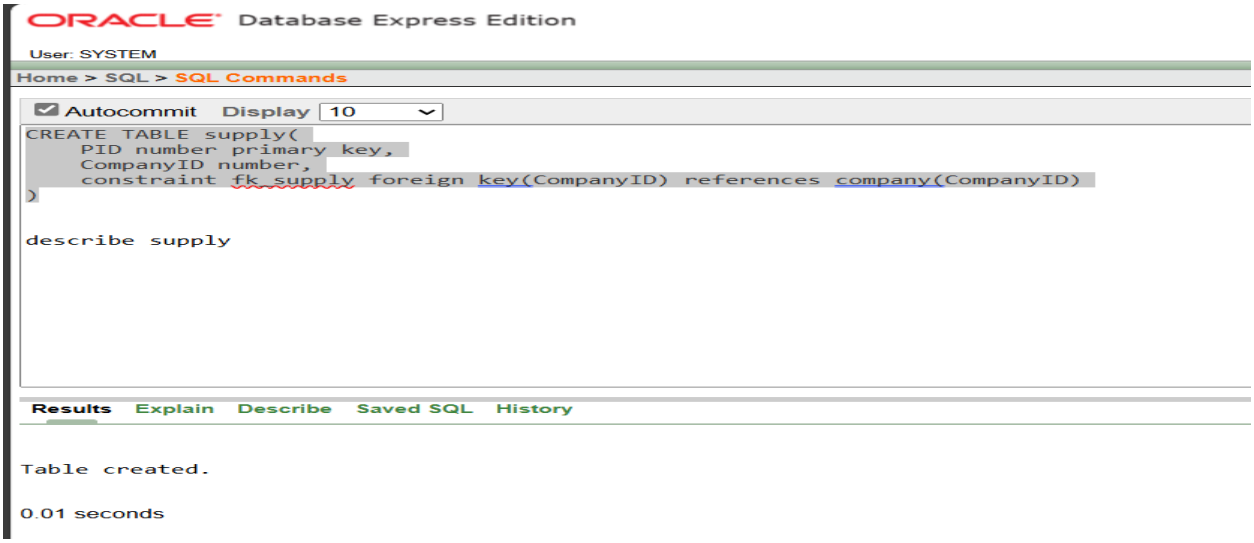


Fig-15: Create supply table

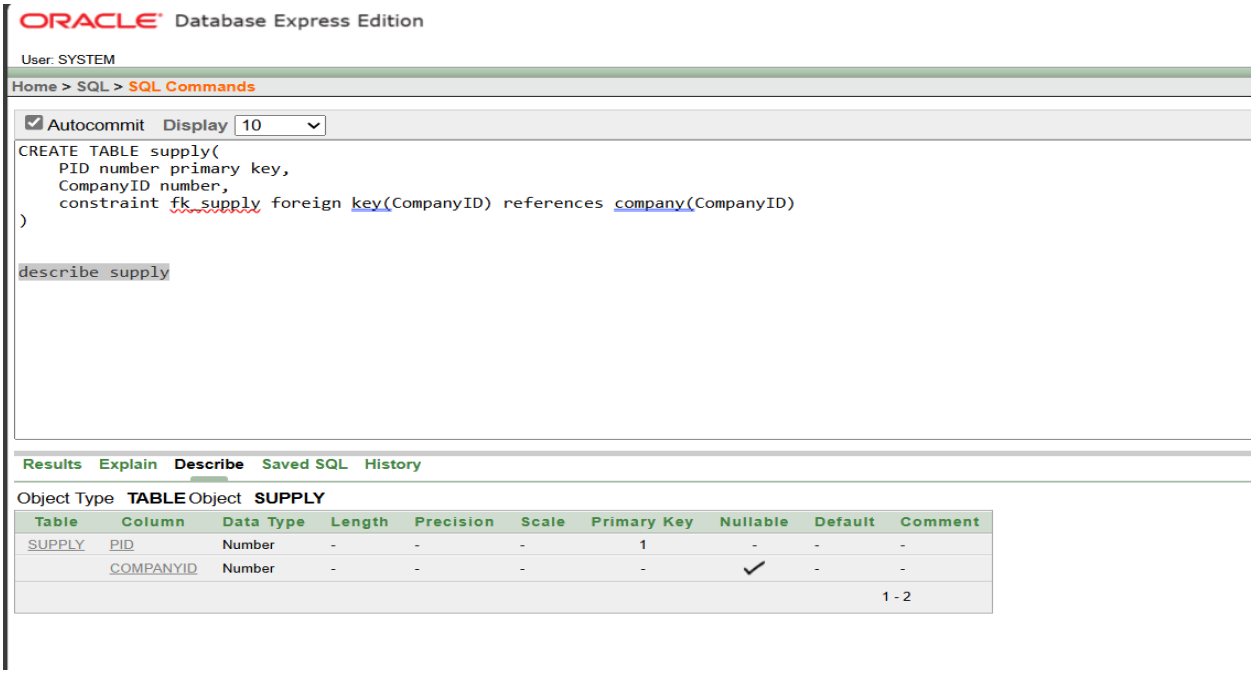


Fig-16: Description of the supply table

9. Category Table

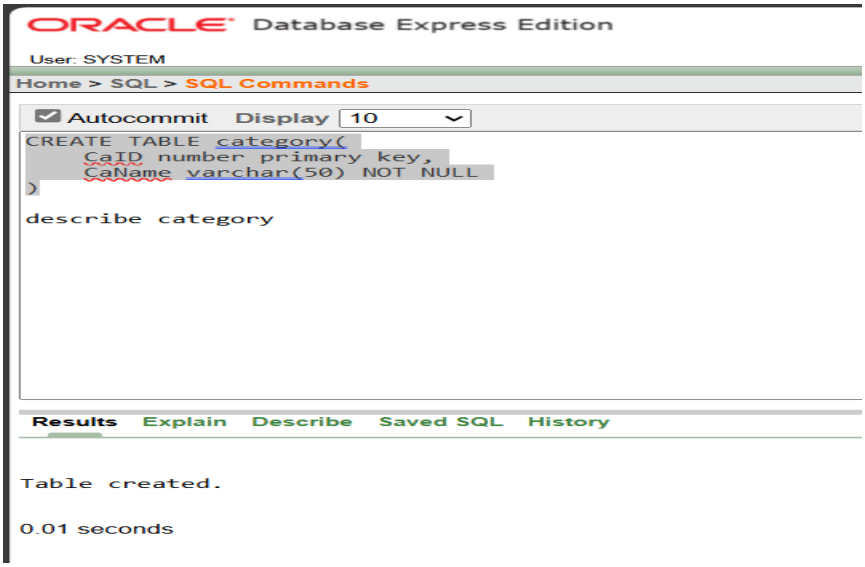


Fig-17: Create category table

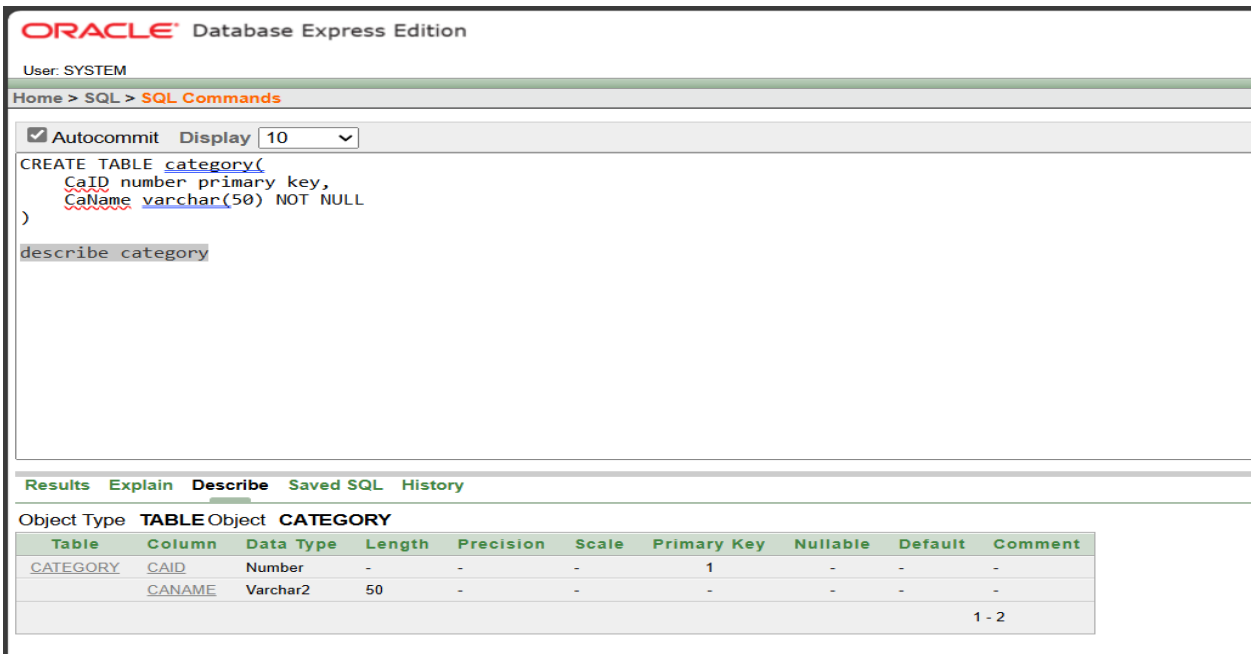


Fig-18: Description of the category table

10. List Table

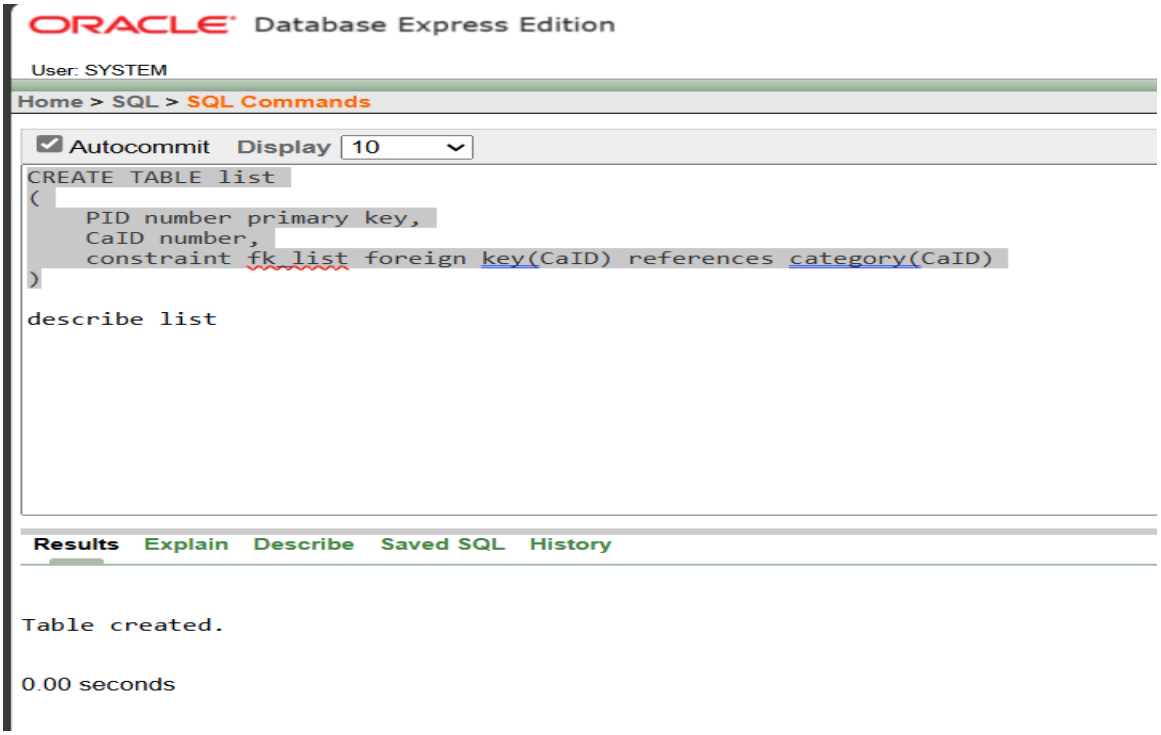


Fig-19: Create list table

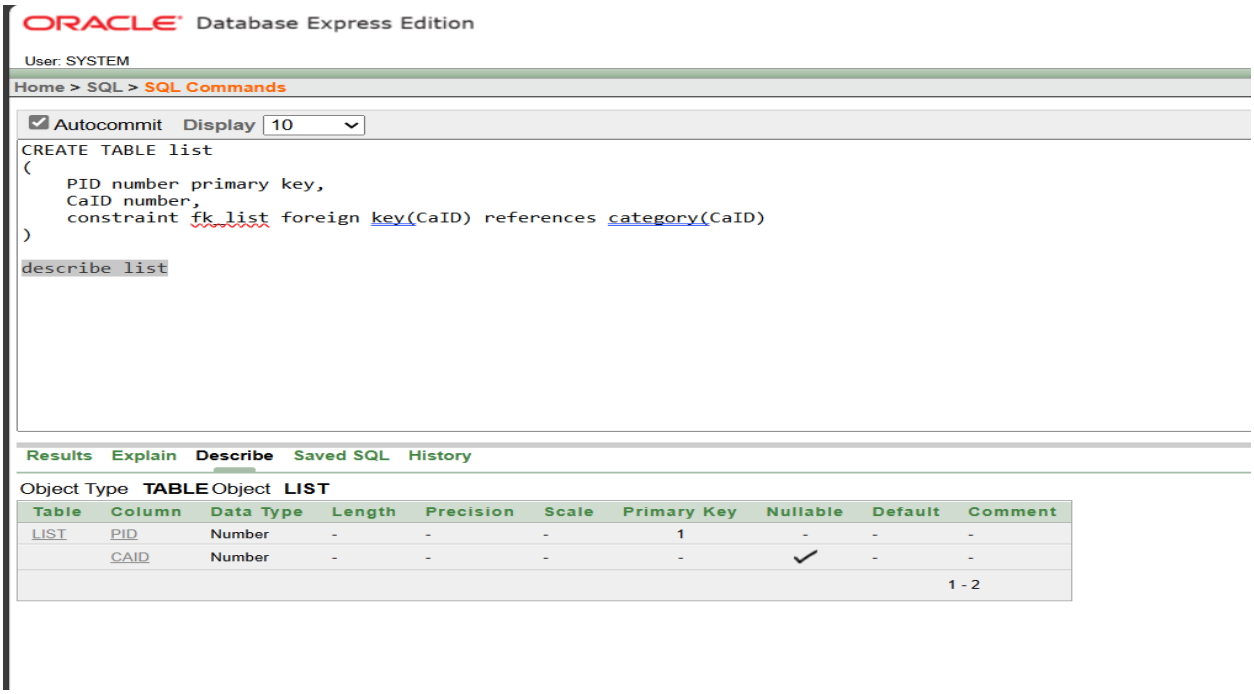


Fig-20: Description of the list table

Inserted Values in the tables

1. Insert into Employee Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 10

```
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1001, 'Taiyeb Tausif Dhrubo', 70000, 23517202);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1002, 'Sanjida Affrin Bristi', 65000, 23517882);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1003, 'Shreya Chakraborty', 60000, 23517562);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1004, 'Montarin Rahman Jousti', 55000, 23526872);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1005, 'Mr. Chowdhury', 50000, 016111111);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1006, 'Ms. Tasnim', 45000, 017222222);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1007, 'Mr. Khan', 40000, 0191111000);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1008, 'Mr. Uddin', 30000, 013001111);
INSERT INTO employee(EID, EName, E_Sal, E_Phn) VALUES (1009, 'Ms. Mina ', 20000, 018000001);


SELECT * FROM employee;
```

Results Explain Describe Saved SQL History

EID	ENAME	E_SAL	E_PHN
1001	Taiyeb Tausif Dhrubo	70000	23517202
1002	Sanjida Affrin Bristi	65000	23517882
1003	Shreya Chakraborty	60000	23517562
1004	Montarin Rahman Jousti	55000	23526872
1005	Mr. Chowdhury	50000	16111111
1006	Ms. Tasnim	45000	17222222
1007	Mr. Khan	40000	191111000
1008	Mr. Uddin	30000	13001111
1009	Ms. Mina	20000	18000001

9 rows returned in 0.00 seconds CSV Export

Fig-21: Inserting values into empolyee table

2. Insert into Customer Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 10

```
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (101, 'Dhrubo', 023517202, 'Gold');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (102, 'Bristy', 023517882, 'Gold');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (103, 'Shreya', 023517562, 'Gold');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (104, 'Jousti', 023516872, 'Gold');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (105, 'Kabir', 012000019, 'Bronze');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (106, 'Rahima', 013440001, 'Bronze');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (107, 'Raju', 014005504, 'Sliver');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (108, 'Tom', 017909300, 'Bronze');
INSERT INTO customer(CID, CName, C_Phn, membership) VALUES (109, 'Jerry', 019789099, 'Gold');

SELECT * FROM customer;
```

Results Explain Describe Saved SQL History

CID	CNAME	C_PHN	MEMBERSHIP
101	Dhrubo	23517202	Gold
102	Bristy	23517882	Gold
103	Shreya	23517562	Gold
104	Jousti	23516872	Gold
105	Kabir	12000019	Bronze
106	Rahima	13440001	Bronze
107	Raju	14005504	Sliver
108	Tom	17909300	Bronze
109	Jerry	19789099	Gold

9 rows returned in 0.00 seconds CSV Export

Fig-22: Inserting values into customer table

7.3. Insert into Product Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 10

```
INSERT INTO product(PID, PName, P_price, quantity) VALUES (1, 'ASUS Laptop', 70000, 10);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (2, 'HP Laptop', 80000, 15);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (3, 'Dell Laptop', 55000, 20);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (4, 'Apple MacBook', 120000, 5);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (5, 'Keyboard', 2000, 30);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (6, 'Printer', 4000, 20);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (7, 'Mouse', 1500, 30);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (8, 'Monitor', 10000, 20);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (9, 'Camera', 70000, 15);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (10, 'Headphone', 1000, 35);
INSERT INTO product(PID, PName, P_price, quantity) VALUES (11, 'Pendrive', 500, 25);


SELECT * FROM product;
```

Results Explain Describe Saved SQL History

PID	PNAME	P_PRICE	QUANTITY
1	ASUS Laptop	70000	10
2	HP Laptop	80000	15
3	Dell Laptop	55000	20
4	Apple MacBook	120000	5
5	Keyboard	2000	30
6	Printer	4000	20
7	Mouse	1500	30
8	Monitor	10000	20
10	Headphone	1000	35
11	Pendrive	500	25

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds CSV Export

Fig- 23: Inserting values into product table

7.4. Insert into Orders Table

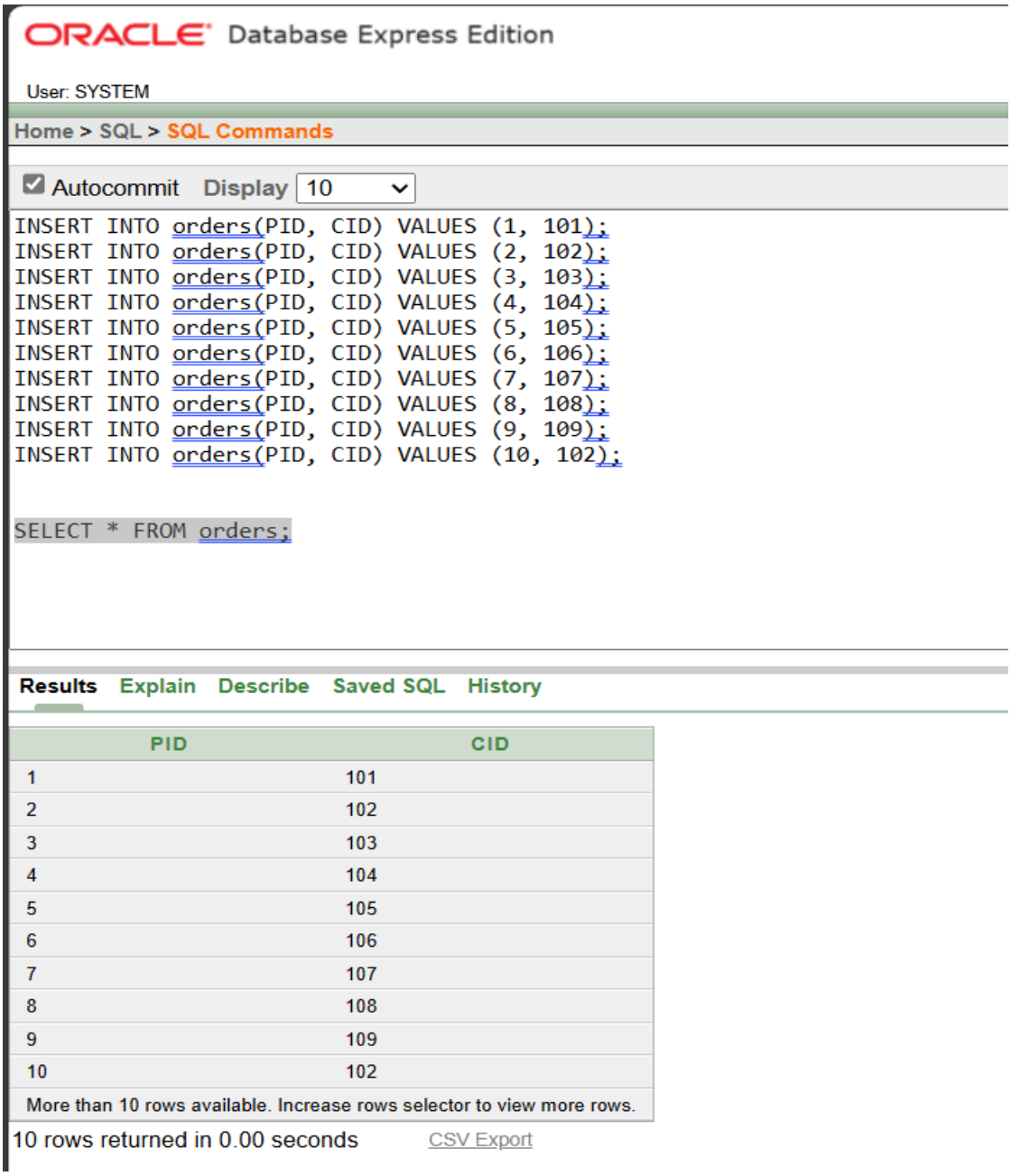


Fig-24: Inserting values into orders table

7.5. Insert into Handle Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 10

```
INSERT INTO handle(CID, CName, membership, EID) VALUES (101, 'Dhrubo','Gold', 1009);
INSERT INTO handle(CID, CName, membership, EID)  VALUES (102, 'Bristy','Gold', 1008);
INSERT INTO handle(CID, CName, membership, EID) VALUES (103, 'Shreya','Gold', 1007);
INSERT INTO handle(CID, CName, membership, EID)  VALUES (104, 'Jousti', 'Gold', 1006);
INSERT INTO handle(CID, CName, membership, EID) VALUES (105, 'Kabir', 'Bronze', 1003);
INSERT INTO handle(CID, CName, membership, EID) VALUES (106, 'Rahima', 'Bronze', 1004);
INSERT INTO handle(CID, CName, membership, EID) VALUES (107, 'Raju', 'Gold', 1001);
INSERT INTO handle(CID, CName, membership, EID) VALUES (108, 'Tom', 'Sliver', 1001);
INSERT INTO handle(CID, CName, membership, EID) VALUES (109, 'Jerry','Gold', 1002);

SELECT * FROM handle;
```

Results Explain Describe Saved SQL History

CID	CNAME	MEMBERSHIP	EID
101	Dhrubo	Gold	1009
102	Bristy	Gold	1008
103	Shreya	Gold	1007
104	Jousti	Gold	1006
105	Kabir	Bronze	1003
106	Rahima	Bronze	1004
107	Raju	Gold	1001
108	Tom	Sliver	1001
109	Jerry	Gold	1002

9 rows returned in 0.00 seconds CSV Export

Fig-25: Inserting values into handle table

6. Insert into Restore Table

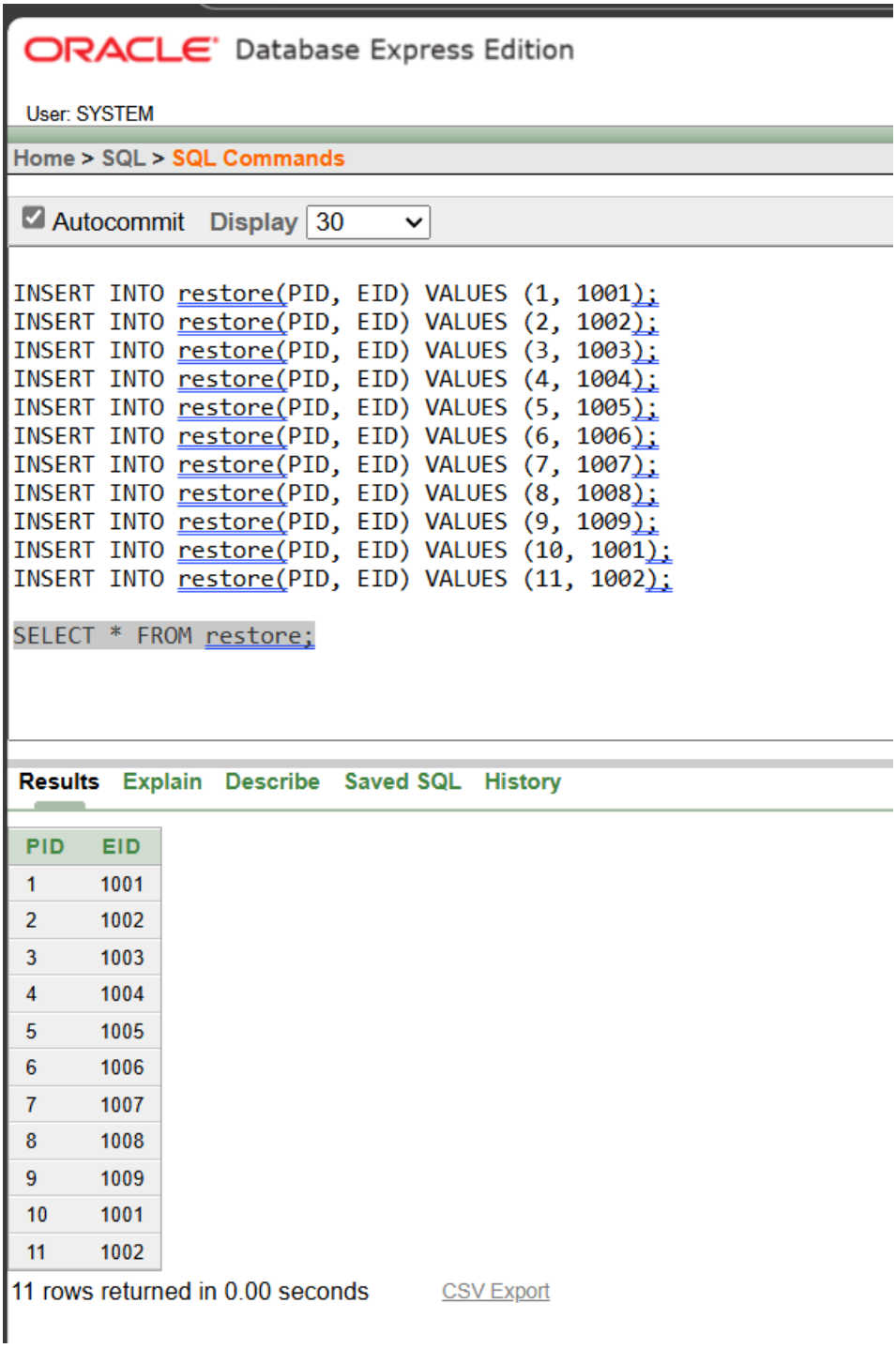


Fig-26: Inserting values into restore table

7. Insert into Company Table

ORACLE Database Express Edition

User: SYSTEM

Home > SQL > SQL Commands

Autocommit Display 30

```
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (100, 'HP', 0023517202);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (101, 'ASUS', 0023517222);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (102, 'Dell', 0023517212);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (103, 'Apple', 0023517112);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (104, 'Canon', 023051799);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (105, '4Tech', 023501700);
INSERT INTO company(CompanyID, CompanyName, CompanyPhn) VALUES (106, 'Logitech', 10235172);

SELECT * FROM company;
```

Results Explain Describe Saved SQL History

COMPANYID	COMPANYNAME	COMPANYPHN
100	HP	23517202
101	ASUS	23517222
102	Dell	23517212
103	Apple	23517112
104	Canon	23051799
105	4Tech	23501700
106	Logitech	10235172

7 rows returned in 0.00 seconds CSV Export

Fig-27: Inserting values into company table

8. Insert into Supply Table

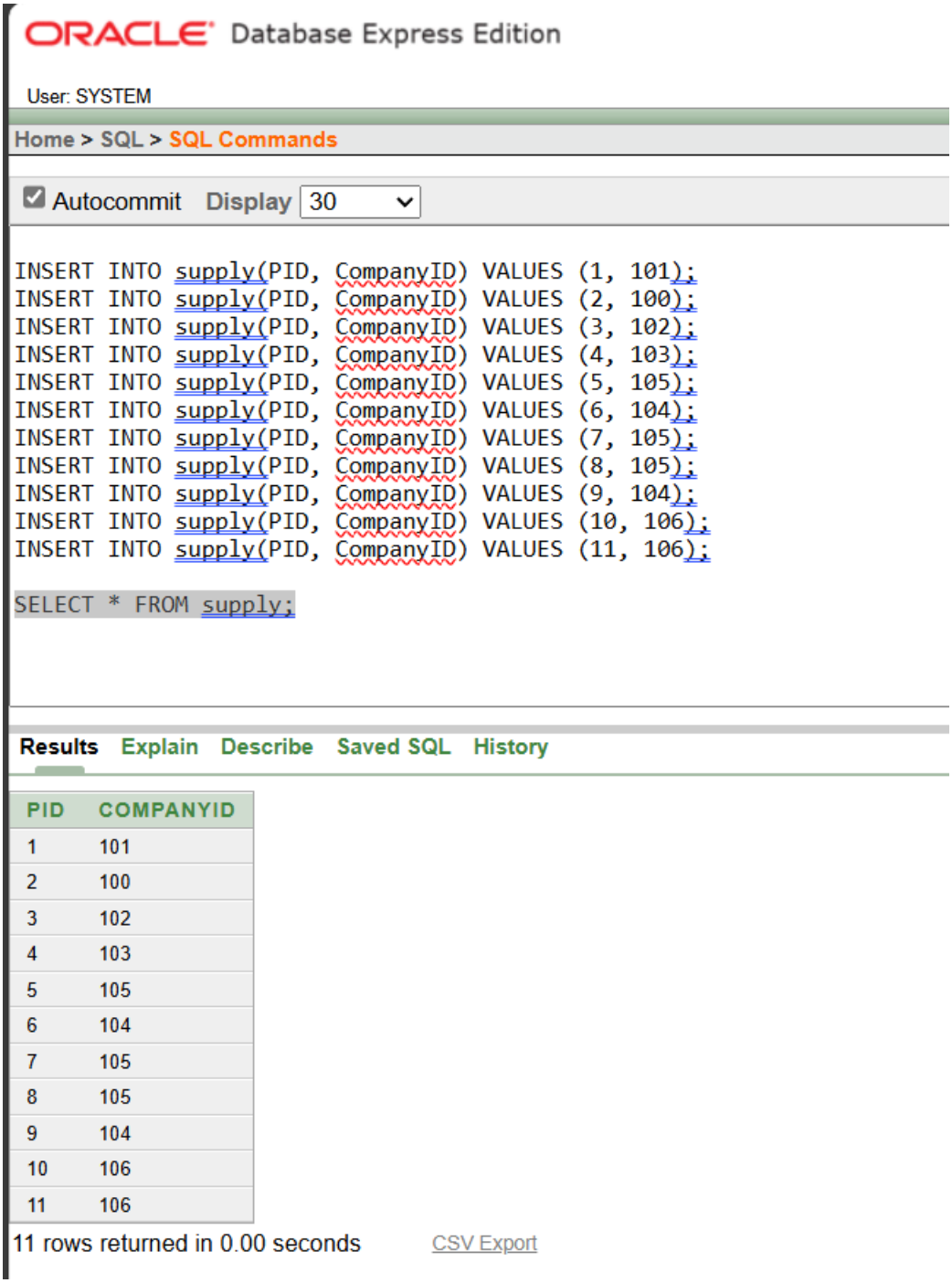


Fig-28: Inserting values into supply table

9. Insert into Category Table

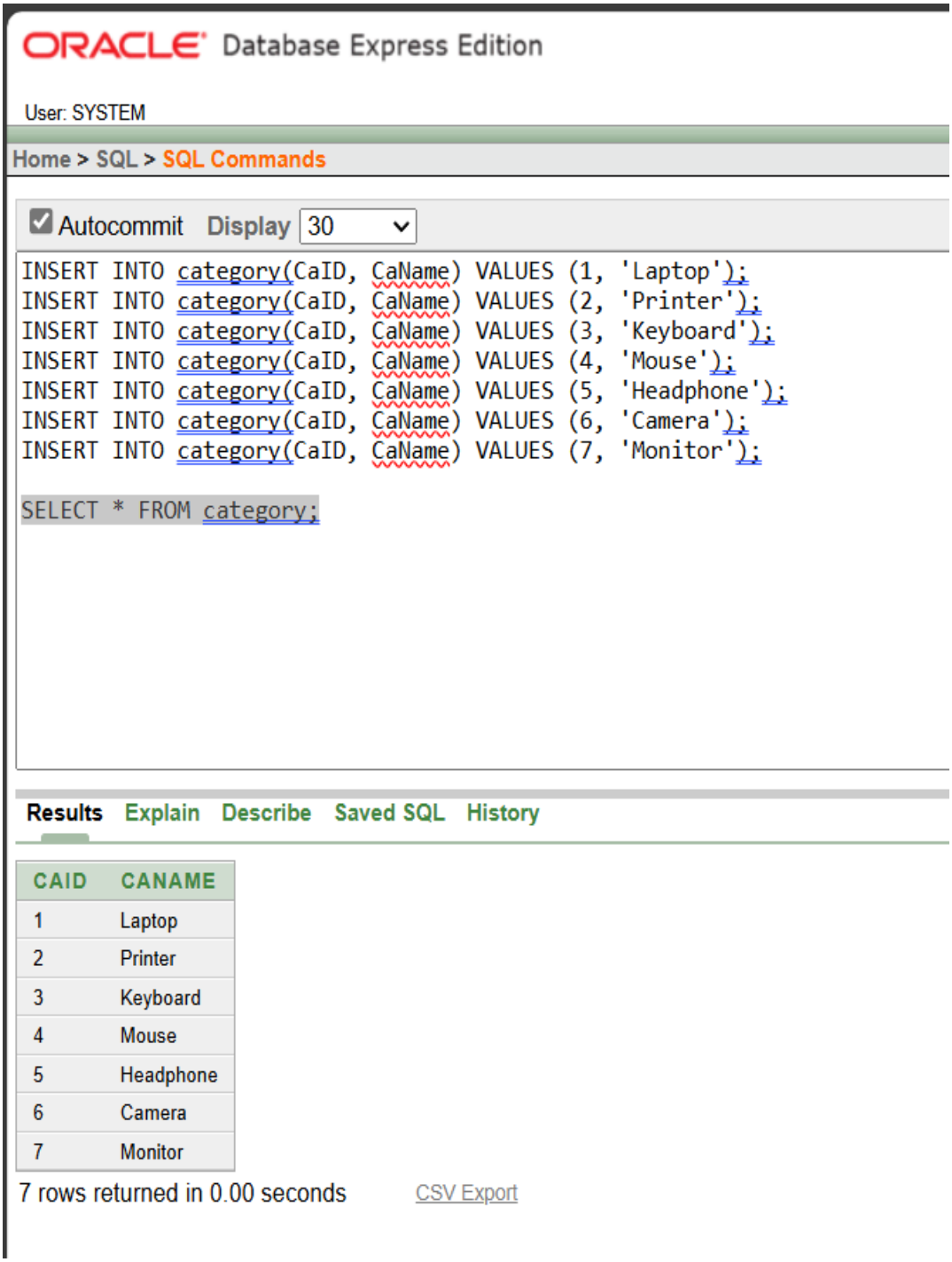


Fig-29: Inserting values into category table

10. Insert into List Table

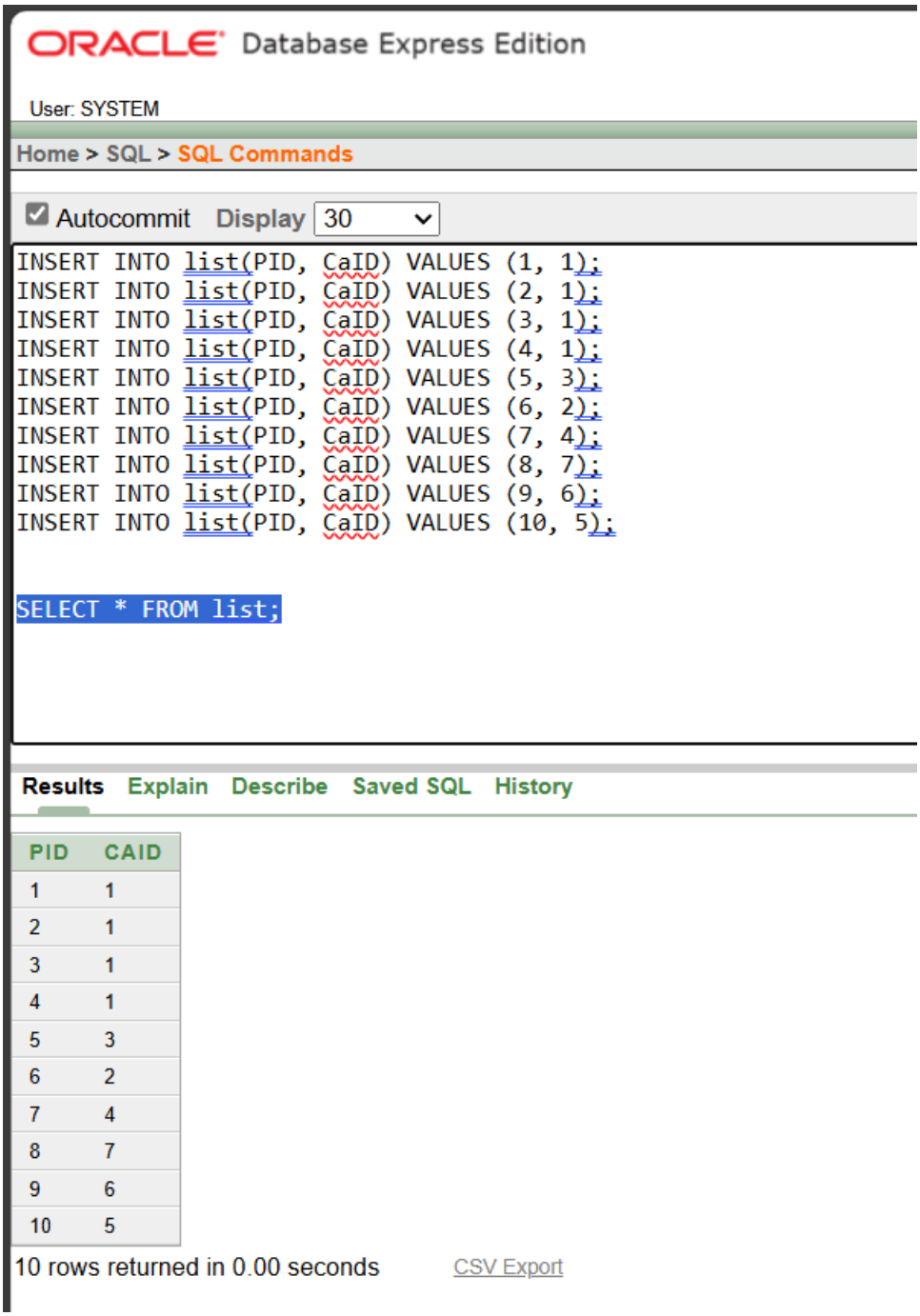


Fig-30: Inserting values into list table

Query Test in DB

a) Simple Query

1. Show product info from the product table.

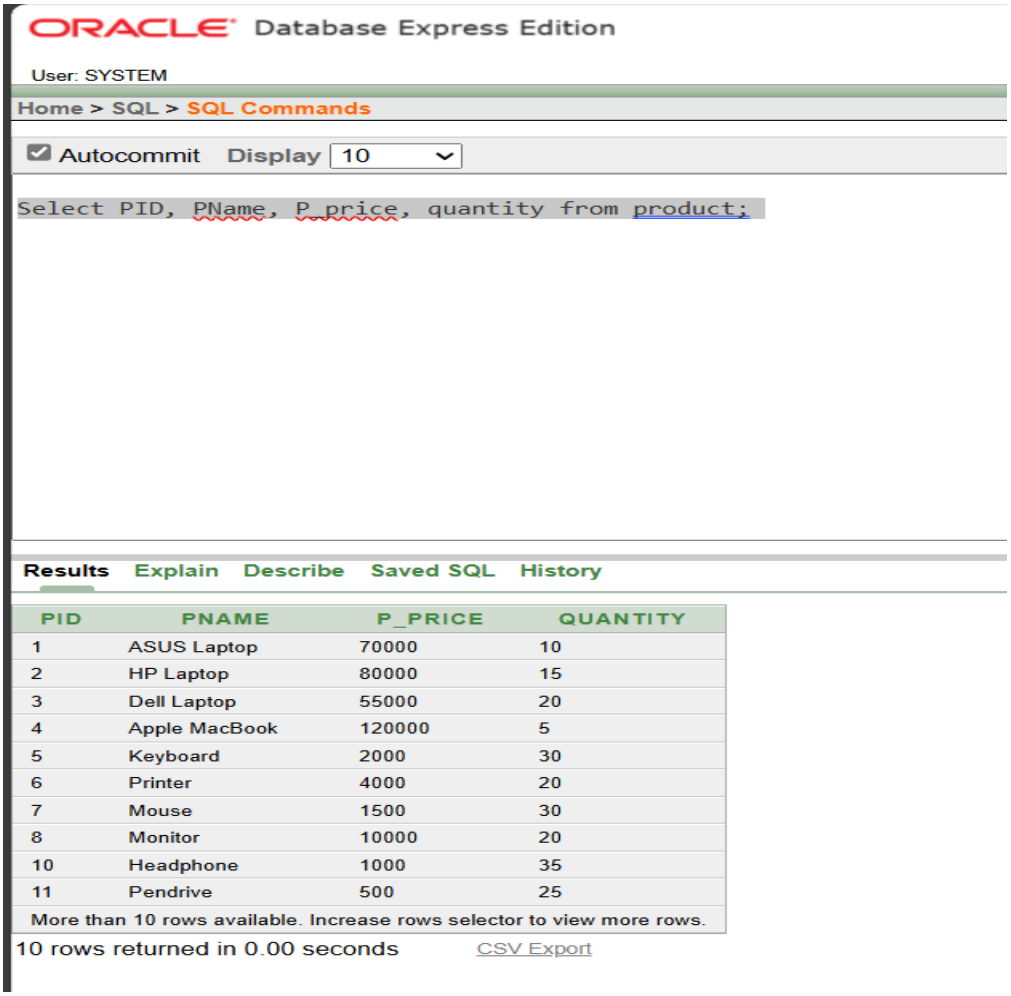


Fig-31: Result of Simple query

2. Show Empolyee ID, Empolyee name, Employee salary where the salary range between 55000 to 70000

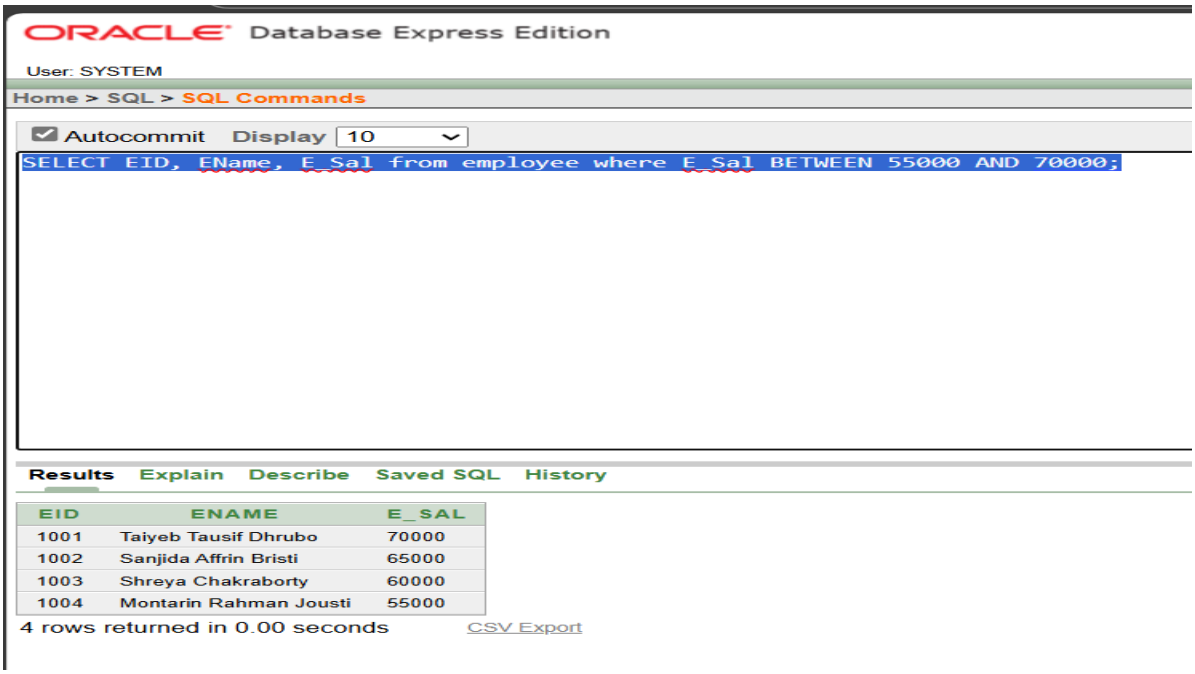


Fig-32 Result of Simple query

b) Query with a single row function

3. Show the customer info which name starts with 'R%' from the customer table

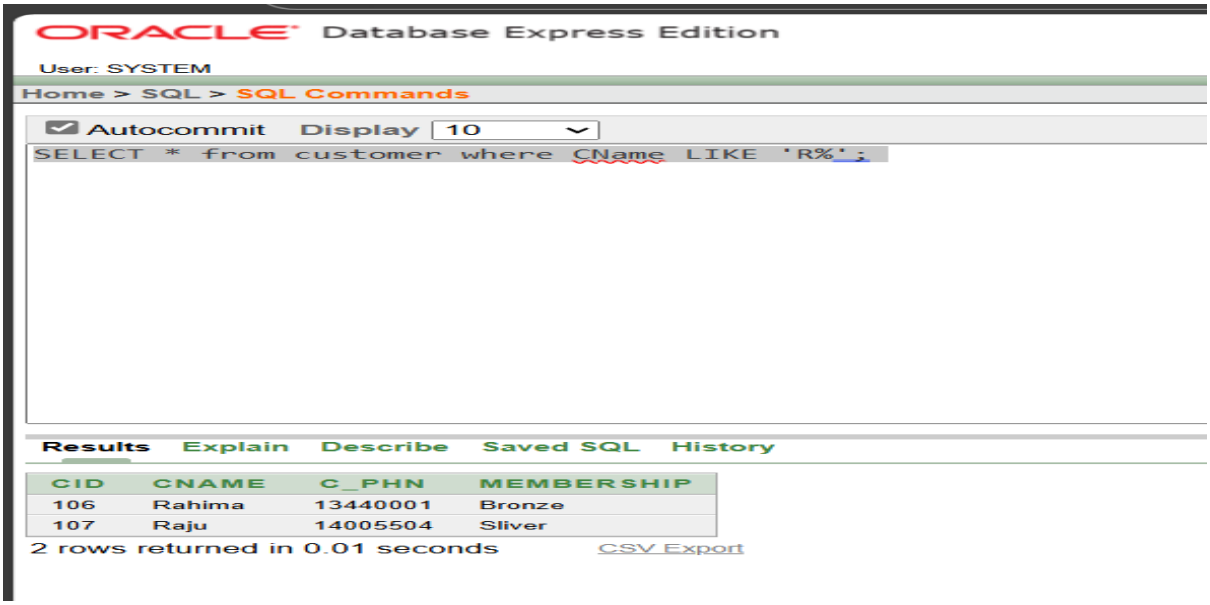


Fig-33: Result of a single row function

c) Query with a Multiple row function/ aggregate function

4. Show the employee info who has lowest salary from the employee table (Aggregate Query)

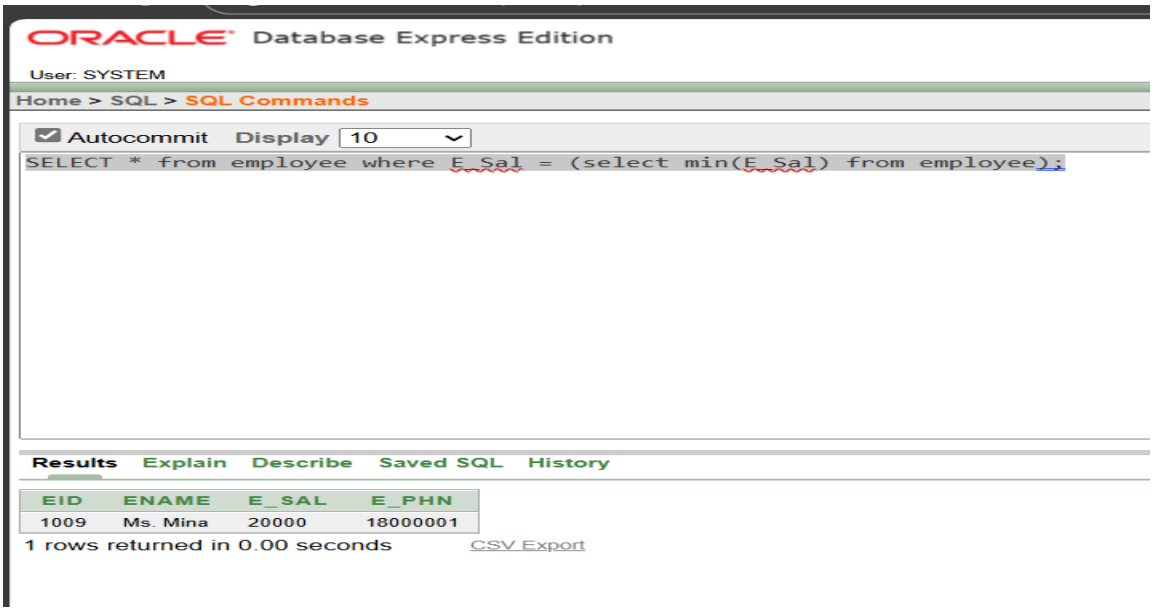


Fig-34: Result of Aggregate query (lowest salary from the employee table)

5. Show the employee info who has maximum salary from the employee table (Aggregate Query)

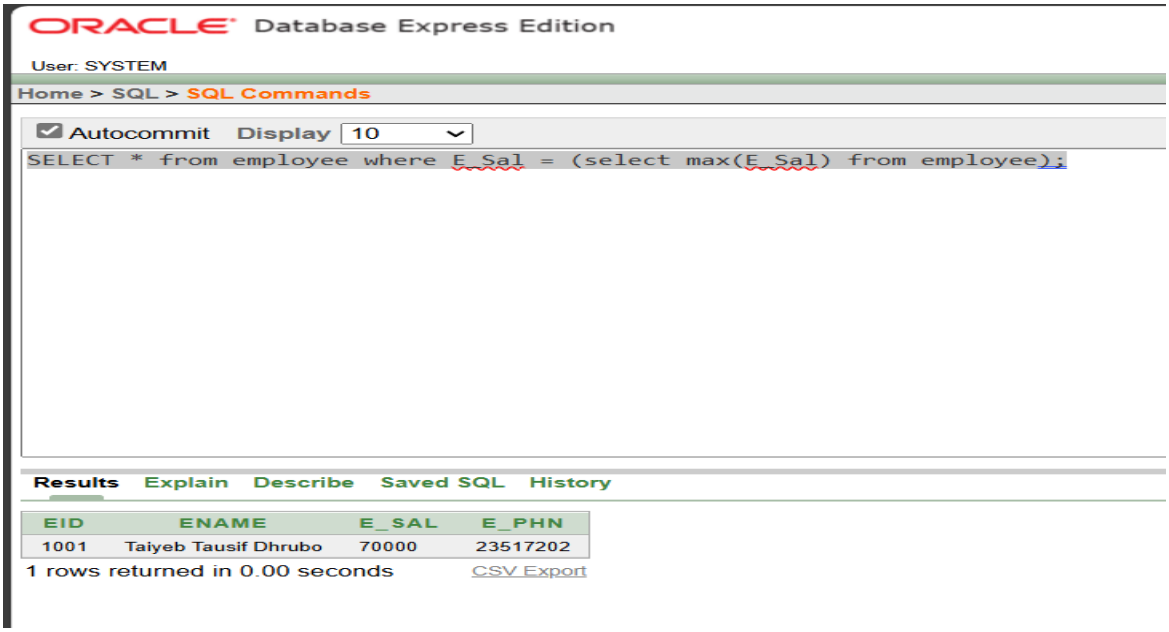


Fig-35: Result of Aggregate query (maximum salary from the employee table)

6. Show total product, overall product quantity from the product table table (Aggregate Query)

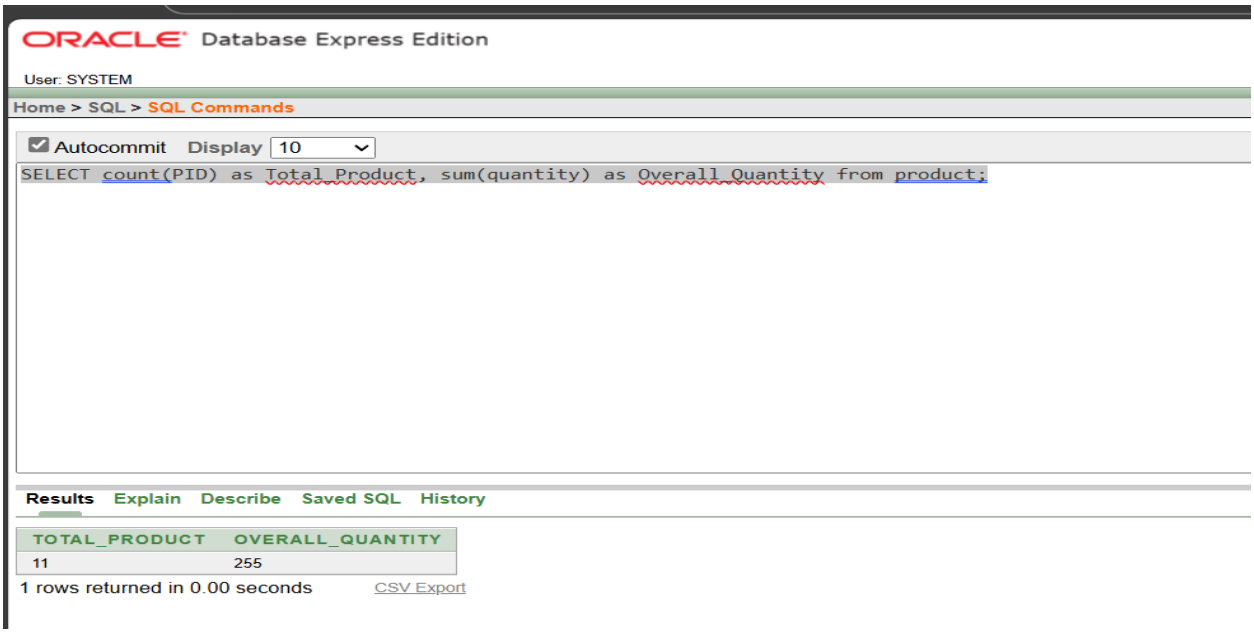


Fig-36: Result of total product, overall product quantity from the product table table (Aggregate Query)

d) Single row subquery

7. Show company ID and Company name of product = 3

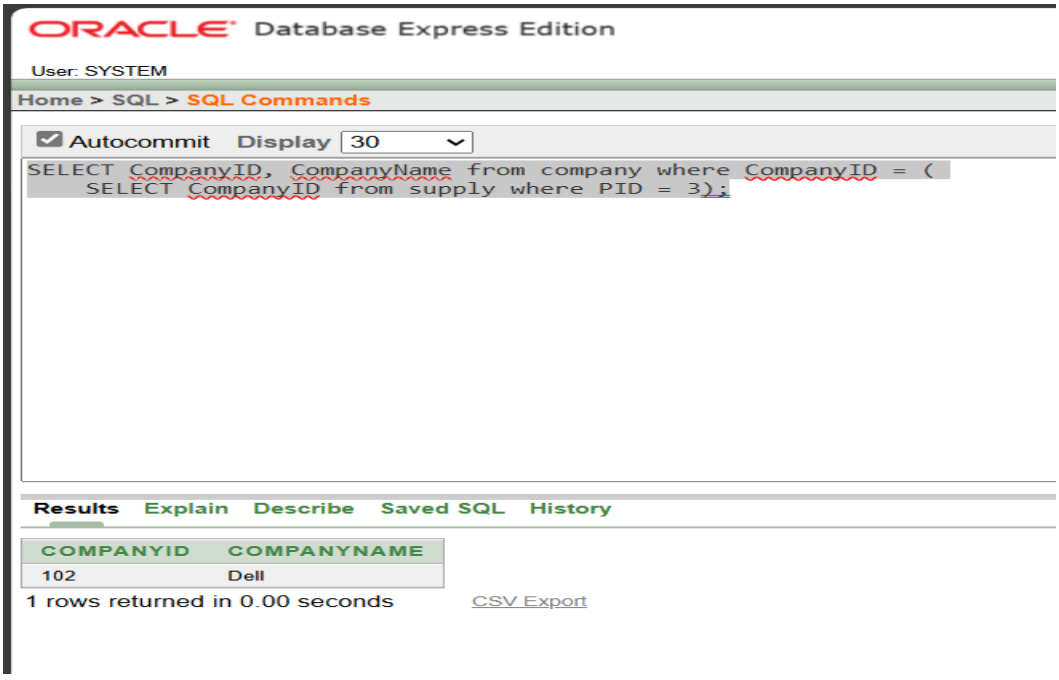


Fig-37: Result of single row subquery (Show company ID and Company name of product = 3)

8. Show the employee info whose salary is greater than Ms. Tasnim (Single Row subquery)

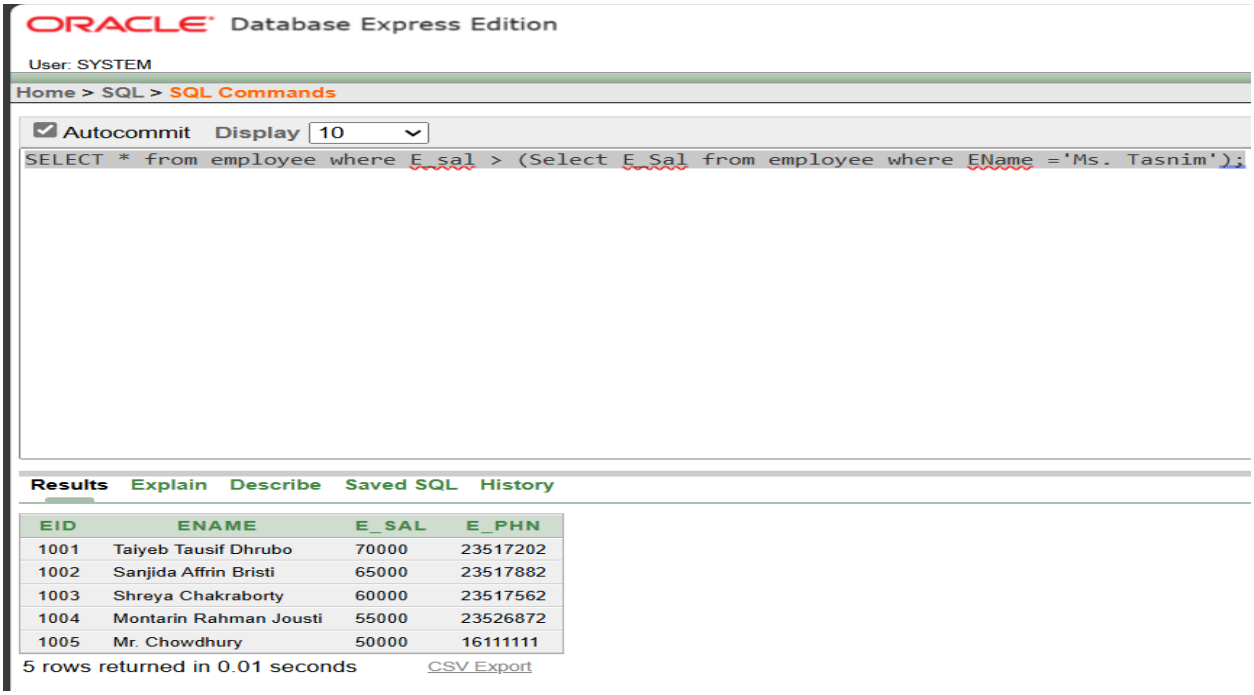


Fig-38: Show the employee info whose salary is greater than Ms. Tasnim (Single Row subquery)

e) Multiple row subquery

9. Show all the details info of the laptop category

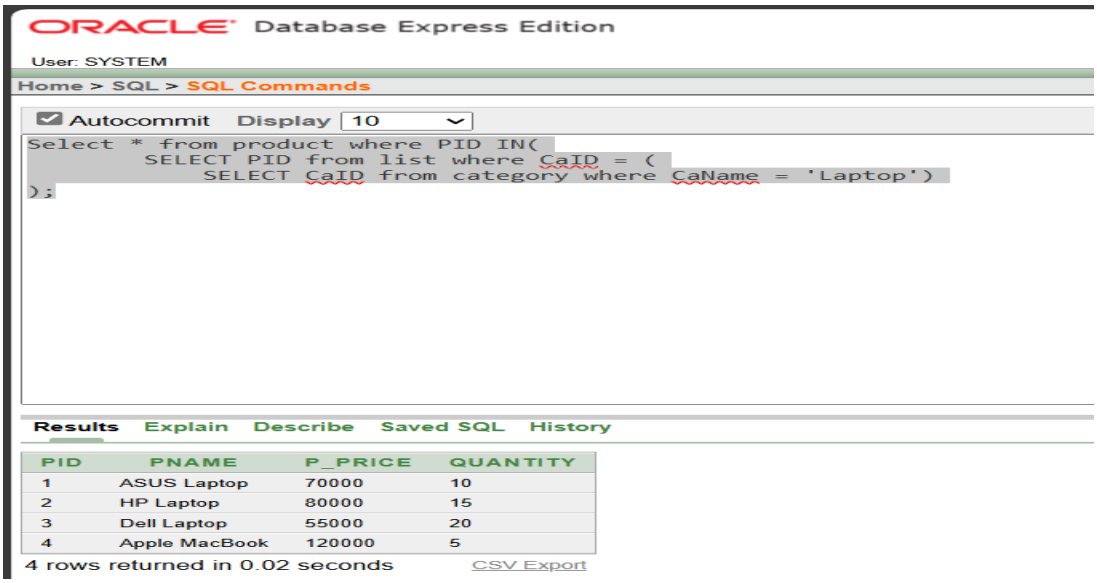


Fig-39: Result of multiple row subquery (Show all the details info of the laptop category)

10. Show the products orders by 'Gold' customers (multiple row subquery)

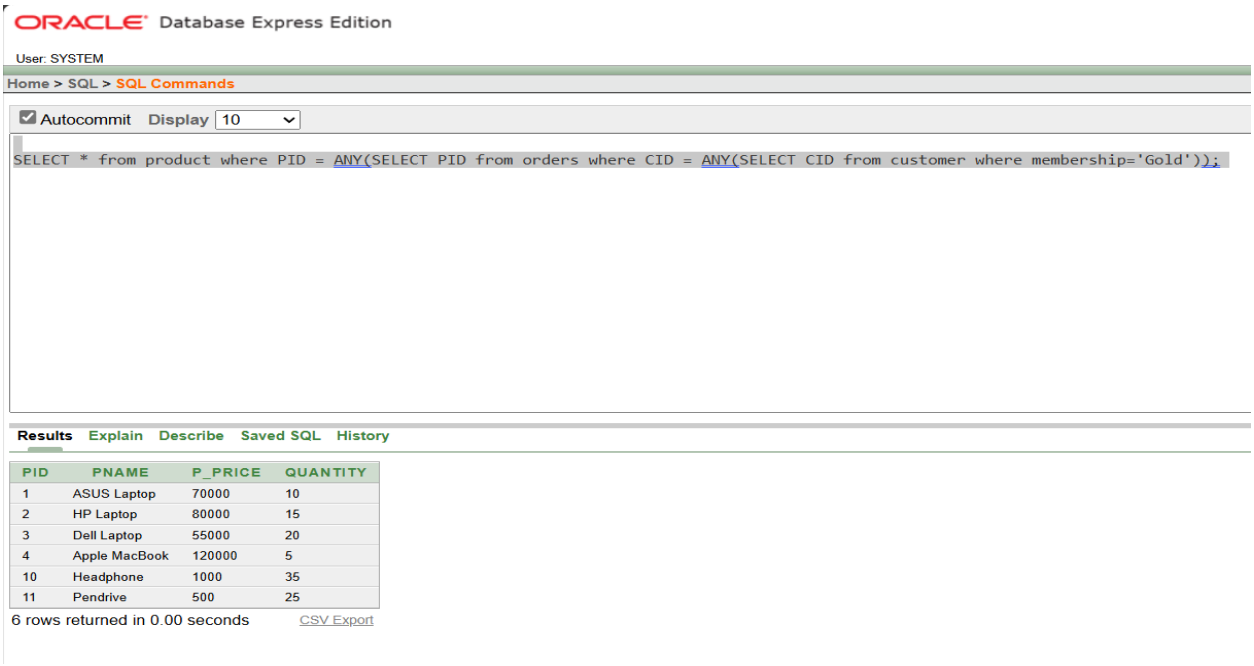


Fig-40: Show the products orders by 'Gold' customers (multiple row subquery)

f) Joining

11. Show the product Name and customer Name who bought the products (Left Join)

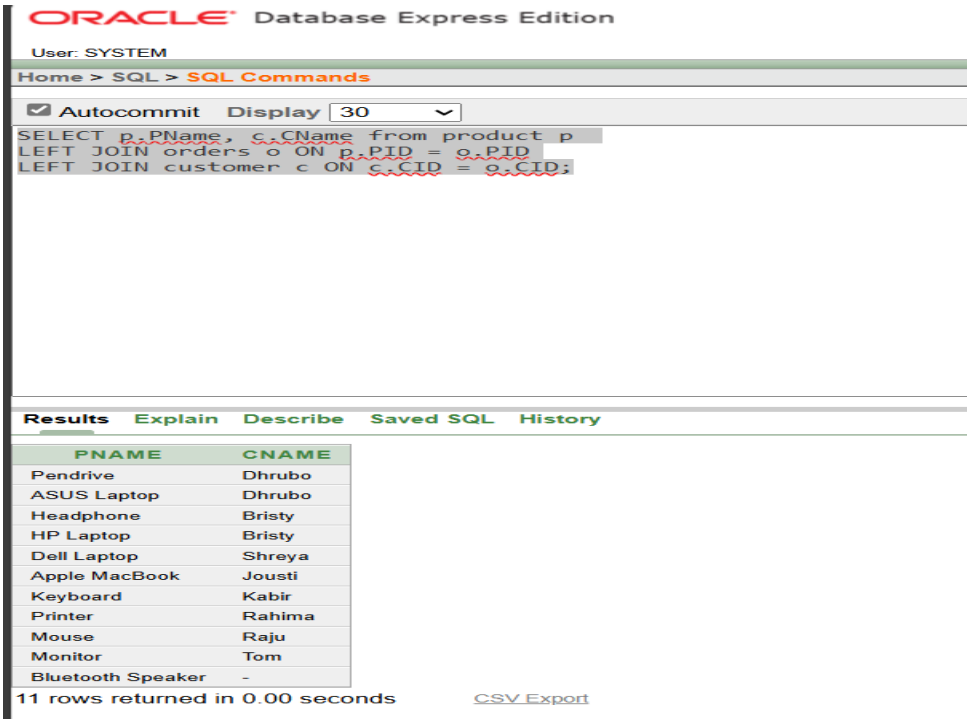


Fig-41: Result of left outer join

12. Show the Customer ID, customer Name and product Name based on Right Join

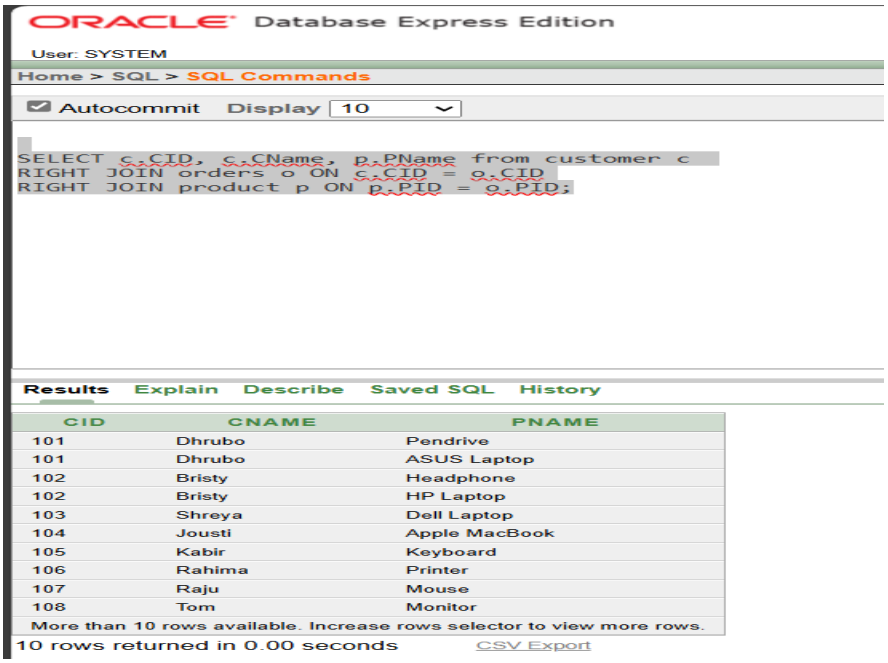


Fig-42: Result of right outer join

13. Show all the product Name and customer Name based on Full Outer Join

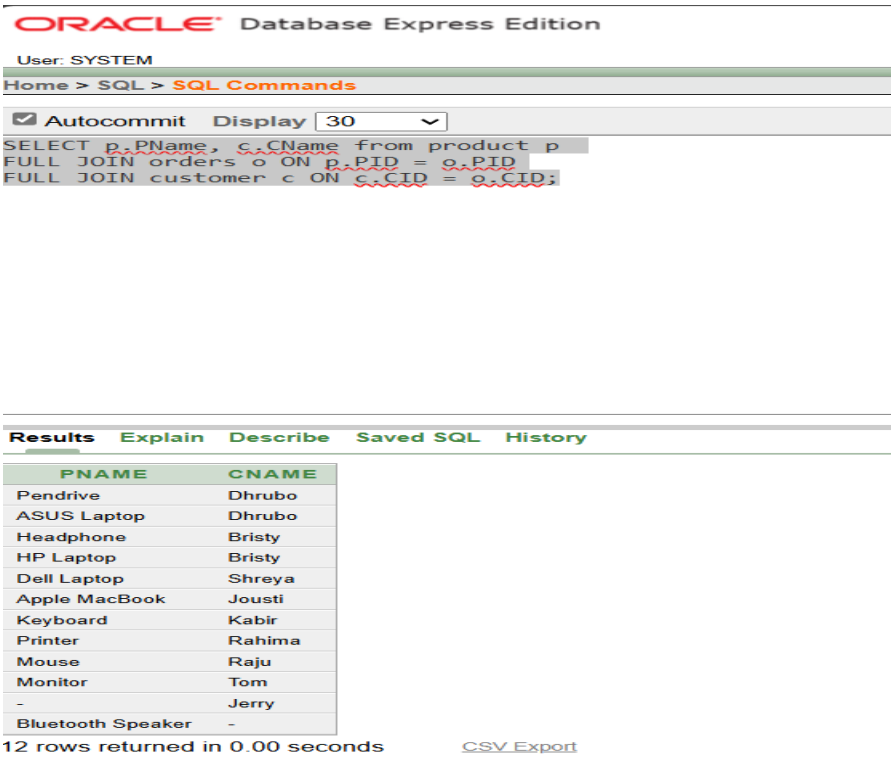


Fig-43: Show all the product Name and customer Name based on Full Outer Join

14. Show all the employee Name, employee salary based on Self Join

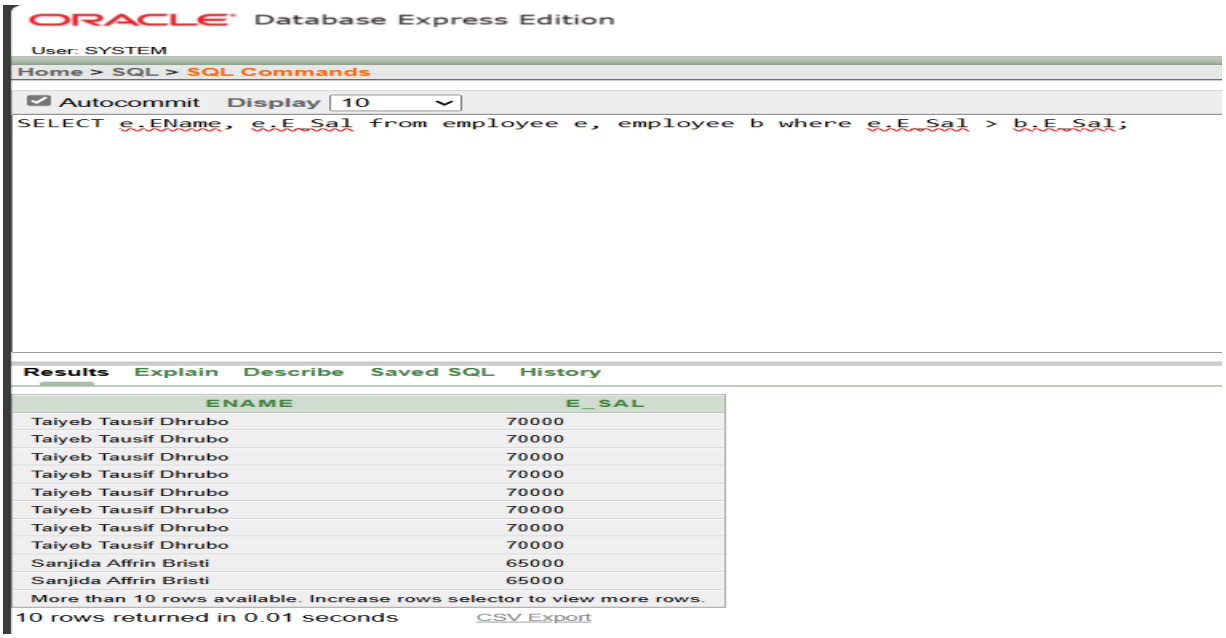


Fig-44: Show all the employee Name, employee salary based on Self Join

15. Show the customer ID, Name, Product ID (Equal Join)

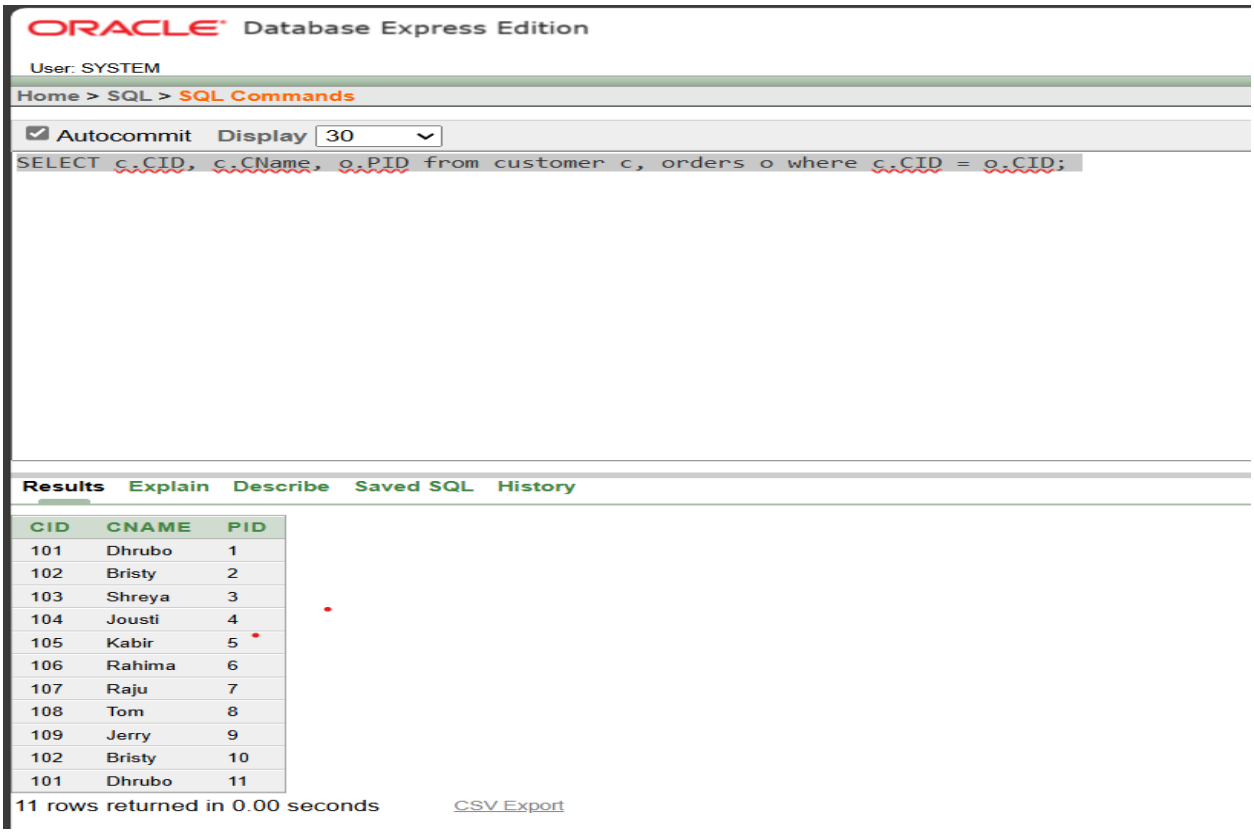


Fig-45: Show the customer ID, Name, Product ID (Equal Join)

16. Show the customer ID, customer Name, membership and product ID based on non-equiJoin

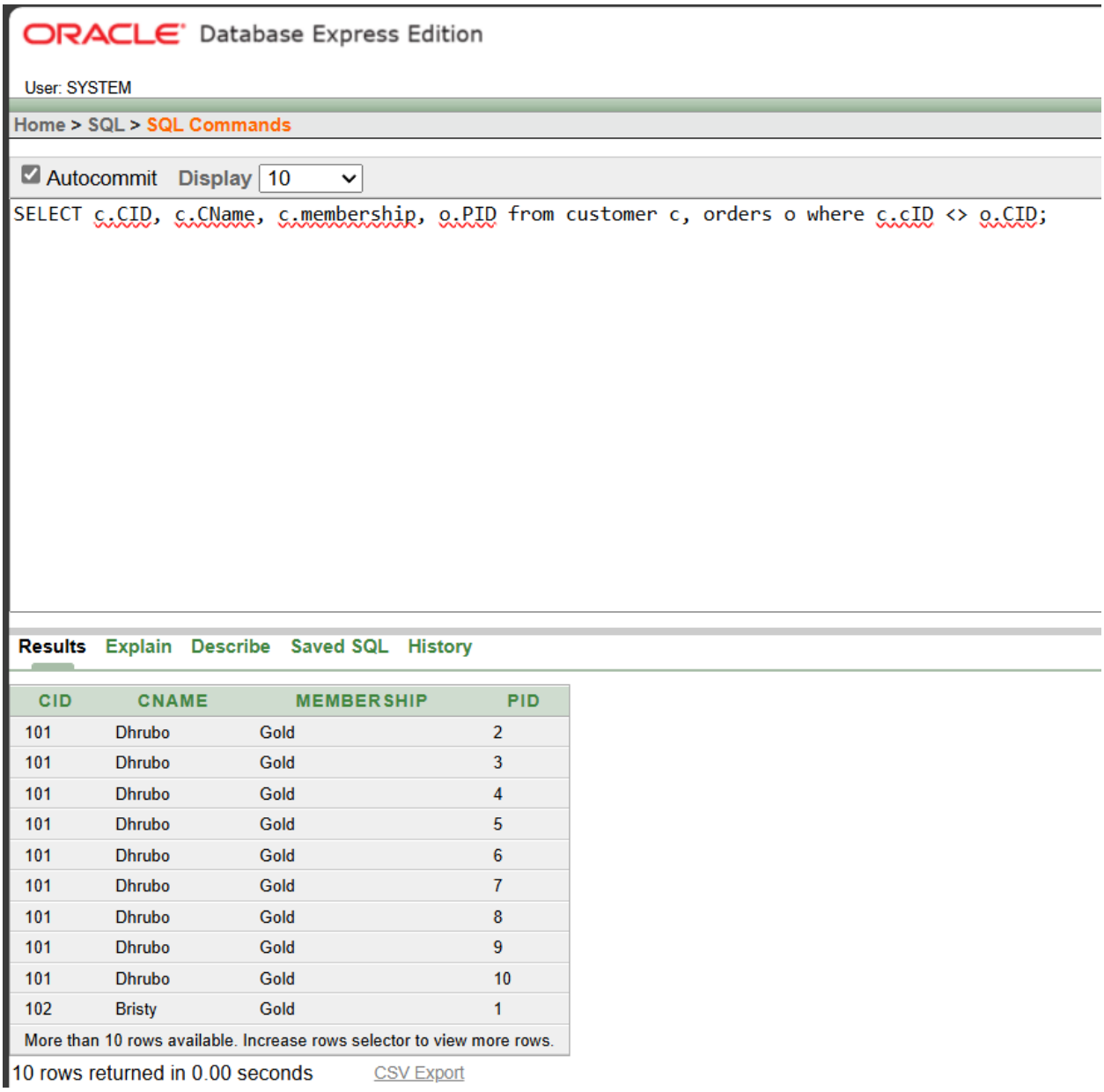


Fig-46: Show the customer ID, customer Name, membership and product ID based on non-equiJoin

g) Create Simple View

17. Create simple view of the company table

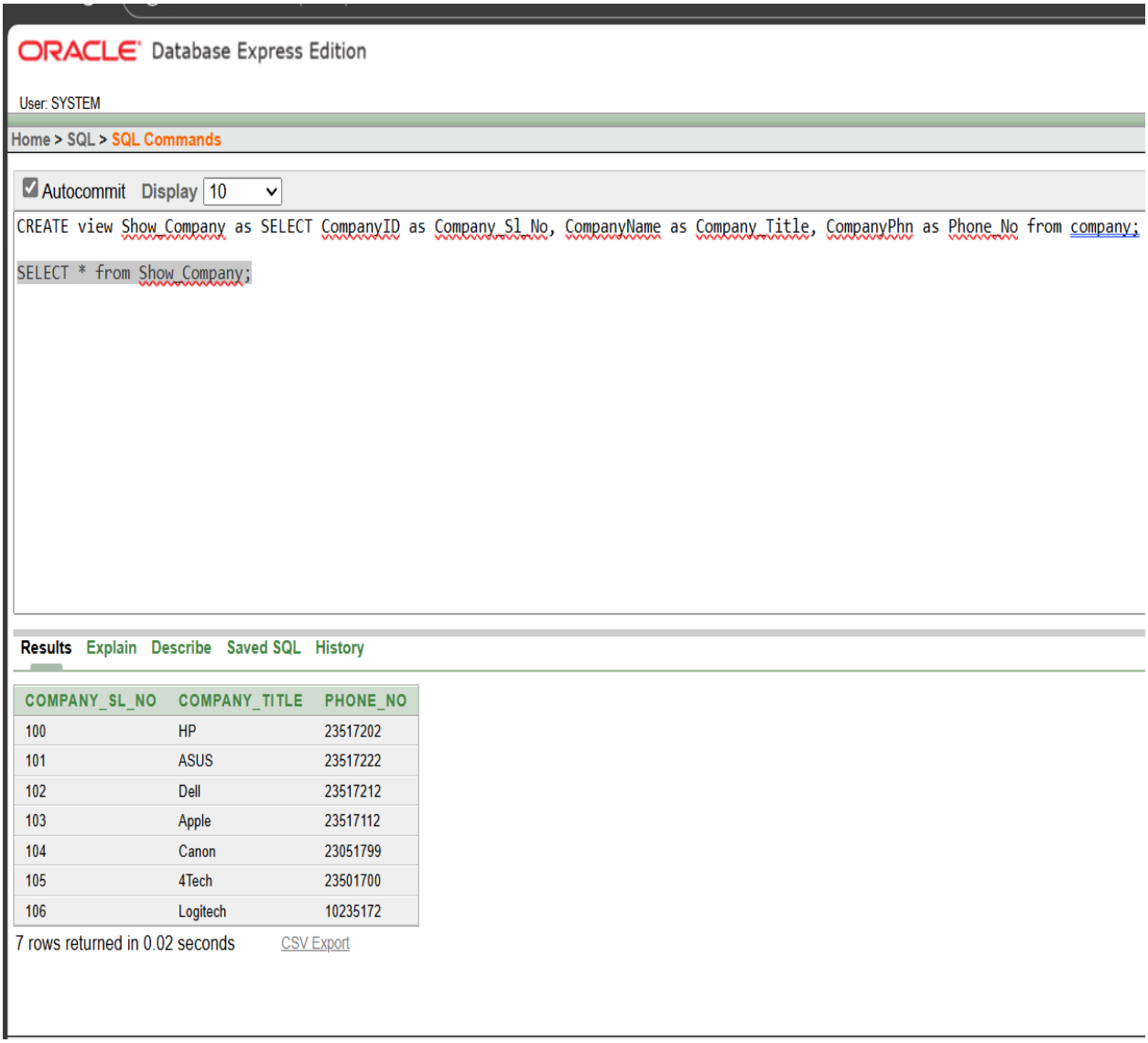


Fig-47: Create simple view of the company table

h) Create Complex View

18. Create Complex View based on restore product list along with employee info

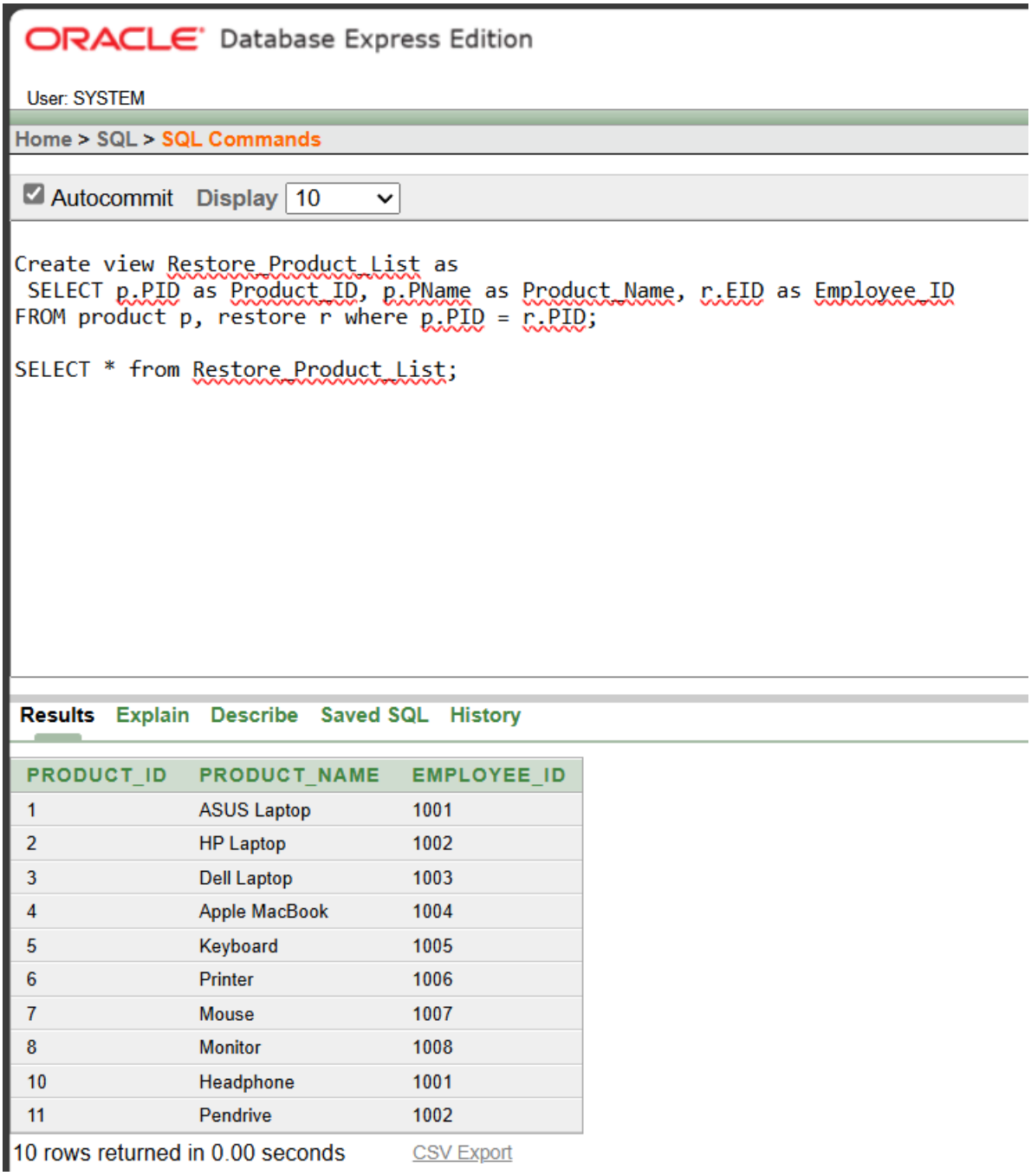


Fig-48: Create Complex View based on restore product list along with employee info

Description of a Successful DB connection

F:\Fall 2024-2025\shopProject\employee.java - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

employee.java

```java
import java.sql.*;
import java.math.*;

public class employee{
  public static void main(String [] args){
      try{
      Class.forName("oracle.jdbc.OracleDriver");
      Connection con = DriverManager.getConnection("jdbc:oracle:thin:@localhost:1521:XE", "System", "Tiger");
      Statement s = con.createStatement();
      ResultSet rs = s.executeQuery("select * from employee");
      System.out.println("Employee_ID" +"\t\t" +"Employee_Name" +"\t\t"
                         +"Employee_Salary" +"\t\t" +"Employee_Phone" +"\t\t" +"\n" );
      while(rs.next()){
          int EID = Integer.parseInt(rs.getString(1));
          String EName = rs.getString(2);
          BigDecimal E_Sal = rs.getBigDecimal(3);
          long E_Phn = rs.getLong(4);

          System.out.print(EID + "\t\t"+EName +"\t\t\t"+E_Sal +"\t\t\t"+E_Phn+"\t\t");

          System.out.println("\n");
      }
      rs.close();
      s.close();
      con.close();
      }
      catch(Exception e){
          System.out.println(e);
      }
  }
```

Fig-49: JDBC code for employee table

```
F:\Fall 2024-2025\shopProject>javac -cp .;ojdbc14.jar employee.java

F:\Fall 2024-2025\shopProject>java -cp .;ojdbc14.jar employee
Employee_ID             Employee_Name           Employee_Salary         Employee_Phone

1001            Taiyeb Tausif Dhrubo                    70000                   23517202

1002            Sanjida Affrin Bristi                   65000                   23517882

1003            Shreya Chakraborty                      60000                   23517562

1004            Montarin Rahman Jousti                  55000                   23526872

1005            Mr. Chowdhury                   50000                   16111111

1006            Ms. Tasnim                      45000                   17222222

1007            Mr. Khan                        40000                   191111000

1008            Mr. Uddin                       30000                   13001111

1009            Ms. Mina                        20000                   18000001
```

Fig-50: Output from JDBC

9. Conclusion

In conclusion, the development and implementation of the database system for super shop management marks a crucial advancement in improving operational efficiency within the retail sector. Utilizing Oracle 10g Express Edition, we have presented a detailed guide for creating a customized database solution. Looking ahead, future efforts may focus on ongoing optimization, the integration of advanced analytics for predictive insights, and adapting to changing market demands. This project serves as a valuable resource for stakeholders seeking to streamline processes and enhance decision-making in super shop management.